

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

CHAPTER 3

Numerical Sequences and Series

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

How to read this book

Everything set in the main column below is Rudin, verbatim. You can read it straight through and never look at an annotation. The annotation layer sits in two places, both outside his text.

Margin notes. A small superscript number in Rudin's text — like this¹ — keys to a numbered note in the outer margin. Since the book prints in black and white, the *tier* of a note is carried by how its number is set:

¹ *The marker is the only mark added to his page, and it is the same for every tier so his line spacing is undisturbed. The number out here tells you what kind of note it is.*

7

Plain number — unpacking. What the sentence is actually saying, where a formula came from, an omitted algebraic step.

7

Outlined number — going deeper. Context beyond the course: what breaks in general, which later theorem this sets up. Skippable.

7

Reversed number — warning. A specific belief students form here that is false, or a place where Rudin's wording will mislead you.

Interleaved boxes. When a note runs longer than a margin will hold, it becomes a box between Rudin's paragraphs, framed by the same three tiers: a **solid** left rule for unpacking, a **dashed** left rule for going deeper, and rules **top and bottom** with a black tab for a warning.

Every figure is an addition. Rudin's Chapter 1 contains no diagrams at all. Any picture you see here is annotation, drawn to expose geometry that his algebra states without illustrating; each carries a grey italic caption.

Where the text is corrected. The main column follows a retyped edition whose own foreword concedes that fresh typos were probably introduced. Where one is evident, the reading is restored from Rudin's third edition and a warning-tier note records the change — silent emendation would make “verbatim” meaningless.

Structural commentary. Two further boxes carry no mathematics at all. They explain the book rather than its contents — why a chapter exists, why Rudin ordered it as he did, why a section that looks like a detour is not one.

HEAVY FRAME, BLACK TITLE BAR

Opens and closes every chapter: what it is for, how it is organized and why in that order, which parts are load-bearing. Where Rudin has explained his own choices — usually in the Preface — these boxes quote him rather than guess.

THIN FRAME, INLINE HEADING | Opens a section: why this material sits here, and what to hold on to.

Books cited in the notes. Where an annotation says another treatment does something differently, it means one of these. They are named because seeing the same theorem set up two ways is often what makes Rudin’s choice legible.

- | | |
|-------------------------------|--|
| <i>Tao, Analysis I–II</i> | Terence Tao, 3rd ed. Builds $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from the Peano axioms and constructs $\mathbb{R}$ by Cauchy sequences rather than cuts. Volume II covers metric spaces and uniform convergence — Rudin’s Chapters 2, 4 and 7. |
| <i>Yu Pin, Tsinghua notes</i> | 数学分析讲义, Tsinghua / Yau Mathematical Sciences Center. Axiomatizes $\mathbb{R}$ by field, order, archimedean and <i>nested interval</i> axioms, then derives the supremum principle. |
| <i>ECNU text</i> | 数学分析 (Mathematical Analysis), 5th ed., 2018, East China Normal University. Standard Chinese two-volume text. Its §7.1 collects six equivalent formulations of completeness and proves them interderivable. |
| <i>USTC notes</i> | 中国科学技术大学少年班数学分析讲义 (程艺 et al., 2005). Teaches the main thread of calculus first and defers real-number theory to a later volume; lists Rudin among its three principal references. |

Chapter 3

NUMERICAL SEQUENCES AND SERIES

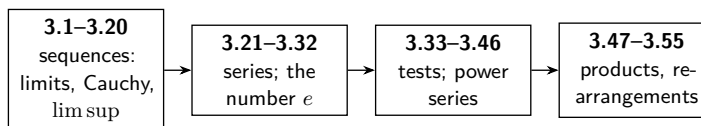
WHY THIS CHAPTER EXISTS

Chapter 1 promised that limits would have somewhere to arrive; Chapter 2 built the spaces they arrive in. This chapter finally defines the arrival — $p_n \rightarrow p$ — and then spends its second half on the single most important special case: series, which are sequences of partial sums.

Two theorems are the chapter's spine. **3.11** says that in $\mathbb{R}^k$ a sequence converges exactly when it is a Cauchy sequence — convergence becomes an *internal* property, checkable without knowing the limit. **3.14** says a monotonic sequence converges exactly when it is bounded. Both are completeness in action, and nearly every convergence test for series (3.24 through 3.44) is one of them wearing different clothes.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin's own account, from the opening paragraph: the chapter “will deal primarily with sequences and series of complex numbers,” but “the basic facts about convergence are just as easily explained in a more general setting” — so the first three sections work in metric spaces, and the theorems proved there are inherited free by $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{R}^k$. The same economy as 1.13(c) and Chapter 2.



Three things to know in advance.

The sequence sections are the load-bearing wall.

Everything about series is a statement about the sequence of partial sums, so 3.21–3.55 consume 3.1–3.20 continuously. In particular $\limsup$ (3.16–3.17), the chapter's hardest definition, exists mainly so that the root test (3.33) and the radius of convergence (3.39) can be stated without exceptions.

Completeness gets its name here. Definition 3.12 finally introduces the word *complete* for “every Cauchy sequence

converges,” discharging the promise made in the comparison box at Definition 1.10. Two more of the ECNU text’s six equivalent completeness statements appear as theorems in this chapter: monotone convergence is 3.14, and the Cauchy criterion is 3.11(c).

The last section is a warning. Rearranging a series can change its sum (3.54) unless the convergence is absolute (3.55). The chapter is arranged so that this lands last, as the reason the distinction of 3.45–3.46 was worth drawing at all.

As the title indicates, this chapter will deal primarily with sequences and series of complex numbers. The basic facts about convergence, however, are just as easily explained in a more general setting. The first three sections will therefore be concerned with sequences in euclidean spaces, or even in metric spaces.

CONVERGENT SEQUENCES

3.1 Definition. A sequence (p_n) in a metric space X is said to *converge* if there is a point $p \in X$ with the following property: For every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies that $d(p_n, p) < \varepsilon$. (Here d denotes the distance in X .)

In this case, we also say that (p_n) *converges to* p , or that p is the *limit* of (p_n) [see Theorem 3.2(b)], and we write $p_n \rightarrow p$, or

$$\lim_{n \rightarrow \infty} p_n = p.$$

If (p_n) does not converge, it is said to *diverge*.²

It might be well to point out that our definition of “convergent sequence” depends not only on (p_n) but also on X ; for instance, the sequence $(1/n)$ converges in $\mathbb{R}^1$ (to 0), but fails to converge in the set of all positive real numbers [with $d(x, y) = |x - y|$]. In cases of possible ambiguity, we can be more precise and specify “convergent in X ” rather than “convergent.”³

We recall that the set of points p_n ($n = 1, 2, 3, \dots$) is the *range* of (p_n) . The range of a sequence may be a finite set, or it may be infinite. The sequence (p_n) is said to be *bounded* if its range is bounded.

As examples, consider the following sequences of complex numbers (that is, $X = \mathbb{R}^2$):

- (a) If $s_n = 1/n$, then $\lim_{n \rightarrow \infty} s_n = 0$; the range is infinite, and the sequence is bounded.
- (b) If $s_n = n^2$, the sequence (s_n) is unbounded, is divergent, and has infinite range.
- (c) If $s_n = 1 + [(-1)^n/n]$, the sequence (s_n) converges to 1, is bounded, and has infinite range.

² Read the quantifiers in the stated order: ε first, then N . The challenge comes before the response, and N may depend on ε — almost always does. Every limit argument in this book is a game of respond-to- ε , and the bracketed reference to 3.2(b) is Rudin flagging that “the limit” needs the uniqueness proved there before the phrase is legitimate.

³ The same relativity as “open” in 2.29, and the same cure: name the ambient space. The sequence $(1/n)$ has not changed; what changed is whether the space still contains the point it is crowding toward. Completeness (3.12) is exactly the property of spaces in which this failure mode — converging toward a missing point — cannot occur.

4 The four powers of i recur forever, so the sequence visits $\{i, -1, -i, 1\}$ in rotation. It diverges even though every point of its range is visited infinitely often — each of the four points is a subsequential limit (3.5), and 3.2(b) says a convergent sequence can have only one.

- (d) If $s_n = i^n$, the sequence (s_n) is divergent, is bounded, and has finite range.⁴
- (e) If $s_n = 1$ ($n = 1, 2, 3, \dots$), then (s_n) converges to 1, is bounded, and has finite range.

We now summarize some important properties of convergent sequences in metric spaces.

3.2 Theorem. Let (p_n) be a sequence in a metric space X .

- (a) (p_n) converges to $p \in X$ if and only if every neighborhood of p contains p_n for all but finitely many n .
- (b) If $p \in X$, $p' \in X$, and if (p_n) converges to p and to p' , then $p' = p$.
- (c) If (p_n) converges, then (p_n) is bounded.
- (d) If $E \subset X$ and if p is a limit point of E , then there is a sequence (p_n) in E such that $p = \lim_{n \rightarrow \infty} p_n$.

Proof.

- (a) Suppose $p_n \rightarrow p$ and let V be a neighborhood of p . For some $\varepsilon > 0$, the conditions $d(q, p) < \varepsilon$, $q \in X$ imply $q \in V$. Corresponding to this ε , there exists N such that $n \geq N$ implies $d(p_n, p) < \varepsilon$. Thus $n \geq N$ implies $p_n \in V$.

Conversely, suppose every neighborhood of p contains all but finitely many of the p_n . Fix $\varepsilon > 0$ and let V be the set of all $q \in X$ such that $d(p, q) < \varepsilon$. By assumption, there exists N (corresponding to this V) such that $p_n \in V$ if $n \geq N$. Thus $d(p_n, p) < \varepsilon$ if $n \geq N$; hence $p_n \rightarrow p$.

- (b) Let $\varepsilon > 0$ be given. There exist integers N, N' such that

$$\begin{aligned} n \geq N & \text{ implies } d(p_n, p) < \frac{\varepsilon}{2}, \\ n \geq N' & \text{ implies } d(p_n, p') < \frac{\varepsilon}{2}. \end{aligned}$$

Hence if $n \geq \max\{N, N'\}$, we have

$$d(p, p') \leq d(p, p_n) + d(p_n, p') < \varepsilon.$$

Since ε was arbitrary, we conclude that $d(p, p') = 0$.⁵

- (c) Suppose $p_n \rightarrow p$. Then there is an integer N such that $n > N$ implies $d(p_n, p) < 1$. Put

$$r = \max\{1, d(p_1, p), \dots, d(p_N, p)\}.$$

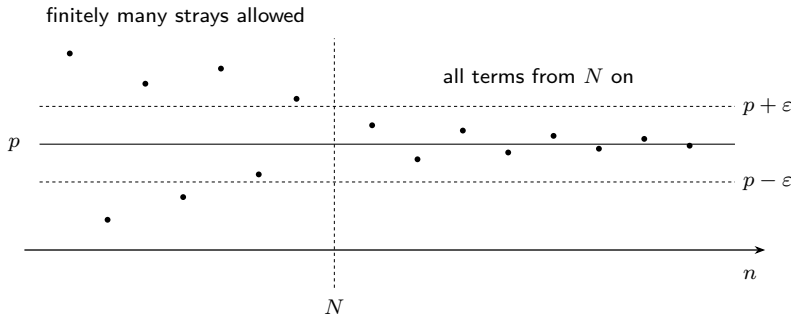
Then $d(p_n, p) \leq r$ for $n = 1, 2, 3, \dots$.⁶

- (d) For each positive integer n , there is a point $p_n \in E$ such that $d(p_n, p) < 1/n$. Given $\varepsilon > 0$, choose N so that $N\varepsilon > 1$. If $n > N$, it follows that $d(p_n, p) < \varepsilon$. Hence $p_n \rightarrow p$.

5 “Since ε was arbitrary” is a standard closing move worth naming: a fixed nonnegative number smaller than every positive ε must be 0 (else take $\varepsilon = d(p, p')/2$). Rudin will use this silently from now on; Exercise 1.5.7 of Chapter 1’s exercises — Tao states it as a lemma — is the same fact.

6 The finite-maximum move of 2.20 and 2.24(c) again, from the other side: the tail is controlled by convergence, and only finitely many early terms remain, so their distances have a maximum. Convergence controls tails; finiteness mops up beginnings.

This completes the proof. $\square$



Convergence, drawn for a real sequence: every ε -tube about p traps the whole tail.

Shrink ε and N moves right; that is the dependence the definition quantifies.

For sequences in $\mathbb{R}^k$ we can study the relation between convergence, on the one hand, and the algebraic operations on the other. We first consider sequences of complex numbers.

3.3 Theorem. Suppose $(s_n), (t_n)$ are complex sequences and

$$\lim_{n \rightarrow \infty} s_n = s, \quad \lim_{n \rightarrow \infty} t_n = t.$$

Then

- (a) $\lim_{n \rightarrow \infty} (s_n + t_n) = s + t$;
- (b) $\lim_{n \rightarrow \infty} cs_n = cs$, $\lim_{n \rightarrow \infty} (c + s_n) = c + s$, for any number c ;
- (c) $\lim_{n \rightarrow \infty} s_n t_n = st$;
- (d) $\lim_{n \rightarrow \infty} (1/s_n) = 1/s$, provided $s_n \neq 0$ ($n = 1, 2, 3, \dots$) and $s \neq 0$.

Proof.

- (a) Given $\varepsilon > 0$, there exist integers N_1, N_2 such that

$$\begin{aligned} n \geq N_1 & \text{ implies } |s_n - s| < \frac{\varepsilon}{2}, \\ n \geq N_2 & \text{ implies } |t_n - t| < \frac{\varepsilon}{2}. \end{aligned}$$

If $N = \max\{N_1, N_2\}$, then $n \geq N$ implies

$$|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon.$$

This proves (a). The proof of (b) is trivial.

- (c) We use the identity

$$s_n t_n - st = (s_n - s)(t_n - t) + s(t_n - t) + t(s_n - s). \quad (3.1)$$

- ⁷ Given $\varepsilon > 0$, there are integers N_1, N_2 such that

$$\begin{aligned} n \geq N_1 & \text{ implies } |s_n - s| < \sqrt{\varepsilon}, \\ n \geq N_2 & \text{ implies } |t_n - t| < \sqrt{\varepsilon}. \end{aligned}$$

⁷ The identity is the whole proof, and it is chosen so that every term on the right involves at least one error factor $s_n - s$ or $t_n - t$. Add and subtract st_n (or $s_n t$) and you find it. The same add-and-subtract manufacture drives the proofs of 4.4, 5.3(b) and the product rules everywhere.

If we take $N = \max\{N_1, N_2\}$, $n \geq N$ implies

$$|(s_n - s)(t_n - t)| < \varepsilon,$$

so that

$$\lim_{n \rightarrow \infty} (s_n - s)(t_n - t) = 0.$$

We now apply (a) and (b) to (3.1) and conclude that

$$\lim_{n \rightarrow \infty} (s_n t_n - st) = 0.$$

(d) Choosing m such that $|s_n - s| < |s|/2$ if $n \geq m$, we see that

$$|s_n| > \frac{1}{2}|s| \quad (n \geq m).$$

8 The standard “stay away from zero” step: once s_n is within $|s|/2$ of s , it cannot be nearer than $|s|/2$ to 0 (reverse triangle inequality, Exercise 1.13). Without it, $1/s_n$ could blow up even with $s \neq 0$. Every quotient-limit proof in the book opens with this move.

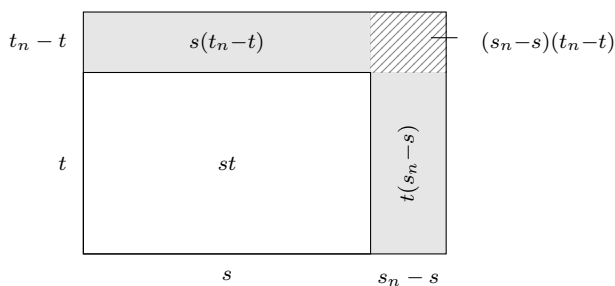
8 Given $\varepsilon > 0$, there is an integer $N > m$ such that $n \geq N$ implies

$$|s_n - s| < \frac{1}{2}|s|^2\varepsilon.$$

Hence, for $n \geq N$,

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s_n - s}{s_n s} \right| < \frac{2}{|s|^2} |s_n - s| < \varepsilon.$$

□



Identity (3.1) as areas: the outer rectangle $s_n t_n$ minus the inner st is two thin strips plus one hatched corner — and each of the three pieces carries at least one error factor. The strips are bounded $\times$ small, the corner is small $\times$ small, which is why $\sqrt{\varepsilon}$ suffices for it. (Drawn for real $s_n > s > 0$, $t_n > t > 0$; the identity is pure algebra and holds for complex sequences.)

HOW THESE PROOFS ARE FOUND, NOT JUST CHECKED

All four parts follow one recipe, and it is the recipe for every ε -argument in the book. Write down the quantity that must be small — $|(s_n + t_n) - (s + t)|$, say. Bound it by pieces you control, each going to 0. Then walk the proof backwards: decide what each piece must be smaller than so the total is under ε , and let that dictate the N 's.

The visible fingerprints of working backwards are the odd-looking

thresholds: $\varepsilon/2$ in (a) because two errors will be added; $\sqrt{\varepsilon}$ in (c) because two errors will be *multiplied*; $\frac{1}{2}|s|^2\varepsilon$ in (d) because the error will be divided by $|s_n s| > \frac{1}{2}|s|^2$. None of these is guessed — each is the final inequality, solved for the intermediate one. This is the same reverse-engineering as the choice of h in 1.21.

3.4 Theorem.

(a) Suppose $\mathbf{x}_n \in \mathbb{R}^k$ ($n = 1, 2, 3, \dots$) and

$$\mathbf{x}_n = (\alpha_{1,n}, \dots, \alpha_{k,n}).$$

Then $(\mathbf{x}_n)$ converges to $\mathbf{x} = (\alpha_1, \dots, \alpha_k)$ if and only if

$$\lim_{n \rightarrow \infty} \alpha_{j,n} = \alpha_j \quad (1 \leq j \leq k). \quad (3.2)$$

(b) Suppose $(\mathbf{x}_n)$, $(\mathbf{y}_n)$ are sequences in $\mathbb{R}^k$, (β_n) is a sequence of real numbers, and $\mathbf{x}_n \rightarrow \mathbf{x}$, $\mathbf{y}_n \rightarrow \mathbf{y}$, $\beta_n \rightarrow \beta$. Then

$$\lim_{n \rightarrow \infty} (\mathbf{x}_n + \mathbf{y}_n) = \mathbf{x} + \mathbf{y}, \quad \lim_{n \rightarrow \infty} \mathbf{x}_n \cdot \mathbf{y}_n = \mathbf{x} \cdot \mathbf{y}, \quad \lim_{n \rightarrow \infty} \beta_n \mathbf{x}_n = \beta \mathbf{x}.$$

Proof.

(a) If $\mathbf{x}_n \rightarrow \mathbf{x}$, the inequalities

$$|\alpha_{j,n} - \alpha_j| \leq |\mathbf{x}_n - \mathbf{x}|,$$

which follow immediately from the definition of the norm in $\mathbb{R}^k$, show that (3.2) holds.

Conversely, if (3.2) holds, then to each $\varepsilon > 0$, there corresponds an integer N such that $n \geq N$ implies

$$|\alpha_{j,n} - \alpha_j| < \frac{\varepsilon}{\sqrt{k}} \quad (1 \leq j \leq k).$$

Hence $n \geq N$ implies

$$|\mathbf{x}_n - \mathbf{x}| = \left\{ \sum_{j=1}^k |\alpha_{j,n} - \alpha_j|^2 \right\}^{1/2} < \varepsilon,$$

so that $\mathbf{x}_n \rightarrow \mathbf{x}$. This proves (a).

Part (b) follows from (a) and Theorem 3.3.⁹

□

⁹ Part (a) is the coordinatewise reduction of 2.39 again: convergence in $\mathbb{R}^k$ is k convergences in $\mathbb{R}$, squeezed between the two norm inequalities $|\alpha_j| \leq |\mathbf{x}| \leq \sqrt{k} \max_j |\alpha_j|$. It is why the rest of the chapter can work with complex or real sequences and lose nothing.

SUBSEQUENCES

WHY SUBSEQUENCES, AND WHY NOW | A subsequence discards terms but keeps order, and the notion pays for itself twice. First, it converts Chapter 2's compactness into a statement about sequences: 3.6(a) is Bolzano–Weierstrass in sequence form, the tool that manufactures limits out of mere boundedness. Second, subsequential limits are the raw material for $\limsup$ and $\liminf$ (3.16), which the series tests will need. Theorem 3.7 — the set of subsequential limits is closed — is stocked for 3.17.

3.5 Definition. Given a sequence (p_n) , consider a sequence (n_k) of positive integers such that $n_1 < n_2 < n_3 < \cdots$. Then the sequence (p_{n_i}) is called a *subsequence* of (p_n) . If (p_{n_i}) converges, its limit is called a *subsequential limit* of (p_n) .

It is clear that (p_n) converges to p if and only if every subsequence of (p_n) converges to p . We leave the details of the proof to the reader.¹⁰

3.6 Theorem.

- (a) If (p_n) is a sequence in a compact metric space X , then some subsequence of (p_n) converges to a point of X .
- (b) Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Proof.

- (a) Let E be the range of (p_n) . If E is finite then there is a $p \in E$ and a sequence (n_i) with $n_1 < n_2 < n_3 < \cdots$ such that

$$p_{n_1} = p_{n_2} = \cdots = p.$$

The subsequence (p_{n_i}) so obtained converges evidently to p .¹¹

If E is infinite, Theorem 2.37 shows that E has a limit point $p \in X$. Choose n_1 so that $d(p, p_{n_1}) < 1$. Having chosen $n_1, \dots, n_{i-1}$, we see from Theorem 2.20 that there is an integer $n_i > n_{i-1}$ such that $d(p, p_{n_i}) < 1/i$. Then (p_{n_i}) converges to p .¹²

- (b) This follows from (a) since Theorem 2.41 implies that every bounded subset of $\mathbb{R}^k$ lies in a compact subset of $\mathbb{R}^k$.

□

3.6(b) is Bolzano–Weierstrass in sequence form

Compare 2.42: every bounded infinite *set* has a limit point. Part (b) is the same fact retold for sequences, and it is the workhorse version — most of analysis manufactures a convergent subsequence from boundedness and then works with it. The ECNU text lists this (致密性定理, its name for sequential compactness) as one of its six equivalent forms of completeness, and Tao takes exactly this property as his *definition* of compactness (his 1.5.1). Note what the proof consumed: 2.37, hence compactness of some k -cell, hence — through 2.40 and 2.38 — the least-upper-bound

10 One direction is trivial; the other is Exercise-level but worth doing once: since $n_k \geq k$ (induction on the strictly increasing indices), the tail of a subsequence is trapped by the tail of the sequence. The statement is used constantly in the contrapositive: to show (p_n) diverges, exhibit two subsequences with different limits — as with i^n in 3.1(d).

11 The finite case is not a formality — it is the pigeonhole: infinitely many indices land on finitely many values, so some value is hit infinitely often. Rudin splits by the range, the set-versus-sequence distinction he flagged at 2.7.

12 Theorem 2.20 — every neighbourhood of a limit point contains infinitely many points of E — is exactly what lets the index increase: within distance $1/i$ there are infinitely many terms, so one of them has index beyond n_{i-1} . The $q \neq p$ clause of 2.18(b) has been earning interest since Chapter 2.

property. Over $\mathbb{Q}$ the statement fails: the sequence $1, 1.4, 1.41, \dots$ of Chapter 1 is bounded with no convergent subsequence in $\mathbb{Q}$.

3.7 Theorem. *The subsequential limits of a sequence (p_n) in a metric space X form a closed subset of X .*

Proof. Let E^* be the set of all subsequential limits of (p_n) and let q be a limit point of E^* . We have to show that $q \in E^*$.

Choose n_1 so that $p_{n_1} \neq q$. (If no such n_1 exists, then E^* has only one point, and there is nothing to prove.) Put $\delta = d(q, p_{n_1})$. Suppose $n_1, \dots, n_{i-1}$ are chosen. Since q is a limit point of E^* , there is an $x \in E^*$ with $d(x, q) < 2^{-i}\delta$. Since $x \in E^*$, there is an $n_i > n_{i-1}$ such that $d(x, p_{n_i}) < 2^{-i}\delta$. Thus

$$d(q, p_{n_i}) \leq 2^{1-i}\delta$$

for $i = 1, 2, 3, \dots$. This says that (p_{n_i}) converges to q . Hence $q \in E^*$.¹³
□

13 A two-step chase: q is approximated by subsequential limits x , and each x is approximated by actual terms p_{n_i} — so terms of the sequence approach q through the middlemen, at the price of a doubled error $2 \cdot 2^{-i}\delta$. The geometric budget $2^{-i}\delta$ is chosen so the doubling still goes to zero. This diagonal-through-approximators pattern returns at 7.23.

CAUCHY SEQUENCES

CONVERGENCE WITHOUT THE LIMIT | Definition 3.1 cannot be checked without naming the limit in advance — a severe handicap, since the whole point of many arguments is to *prove* a limit exists. This section removes the handicap. A Cauchy sequence is one whose terms crowd *each other*; Theorem 3.11 says that in $\mathbb{R}^k$ this internal crowding already guarantees a limit. Rudin says exactly this in the note after 3.11, and it is the single most-used fact in the chapter: every series convergence test works by verifying the Cauchy condition, never by producing the sum.

3.8 Definition. A sequence (p_n) in a metric space X is said to be a *Cauchy sequence* if for every $\varepsilon > 0$ there is an integer N such that $d(p_n, p_m) < \varepsilon$ if $n \geq N$ and $m \geq N$.¹⁴

In our discussion of Cauchy sequences, as well as other situations which will arise later, the following geometric concept will be useful.

3.9 Definition. Let E be a nonempty subset of a metric space X and let S be the set of all real numbers of the form $d(p, q)$ with $p \in E$ and $q \in E$. The sup of S is called the *diameter* of E .

If (p_n) is a sequence in X and if E_N consists of the points $p_N, p_{N+1}, p_{N+2}, \dots$, it is clear from the two preceding definitions that (p_n) is a Cauchy sequence if and only if

$$\lim_{N \rightarrow \infty} \text{diam } E_N = 0.$$

3.10 Theorem.

14 Both indices are quantified, independently. The weaker condition $d(p_n, p_{n+1}) \rightarrow 0$ — consecutive terms approaching — does not suffice: $s_n = \sqrt{n}$ has $s_{n+1} - s_n \rightarrow 0$ yet marches to infinity, and the partial sums of the harmonic series (3.28) do the same. Mistaking the consecutive condition for the Cauchy condition is the most common error made with this definition.

(a) If $\overline{E}$ is the closure of a set E in a metric space X , then

$$\text{diam } \overline{E} = \text{diam } E.$$

(b) If K_n is a sequence of compact sets in X such that $K_n \supset K_{n+1}$ ($n = 1, 2, 3, \dots$) and if

$$\lim_{n \rightarrow \infty} \text{diam } K_n = 0,$$

then $\bigcap_{n=1}^{\infty} K_n$ consists of exactly one point.

Proof.

(a) Since $E \subset \overline{E}$, it is clear that

$$\text{diam } E \leq \text{diam } \overline{E}.$$

Fix $\varepsilon > 0$, and choose $p \in \overline{E}$, $q \in \overline{E}$. By the definition of $\overline{E}$, there are points p', q' in E such that $d(p, p') < \varepsilon$, $d(q, q') < \varepsilon$. Hence

$$\begin{aligned} d(p, q) &\leq d(p, p') + d(p', q') + d(q', q) \\ &< 2\varepsilon + d(p', q') \leq 2\varepsilon + \text{diam } E. \end{aligned}$$

It follows that

$$\text{diam } \overline{E} \leq 2\varepsilon + \text{diam } E,$$

and since ε was arbitrary, (a) is proved.

(b) Put $K = \bigcap_{n=1}^{\infty} K_n$. By Theorem 2.36, K is not empty. If K contains more than one point, then $\text{diam } K > 0$. But for each n , $K_n \supset K$ so that $\text{diam } K_n \geq \text{diam } K$. This contradicts the assumption that $\text{diam } K_n \rightarrow 0$.¹⁵

□

15 Division of labour: 2.36 (compactness) supplies at least one point; shrinking diameters supply at most one. Nested-interval arguments always have these two halves, and it is worth noticing which hypothesis funds which.

3.11 Theorem.

- (a) In any metric space X , every convergent sequence is a Cauchy sequence.
- (b) If X is a compact metric space and if (p_n) is a Cauchy sequence in X , then (p_n) converges to some point of X .
- (c) In $\mathbb{R}^k$, every Cauchy sequence converges.

Note: The difference between the definition of convergence and the definition of a Cauchy sequence is that the limit is explicitly involved in the former, but not in the latter. Thus Theorem 3.11(b) may enable us to decide whether or not a given sequence converges without knowledge of the limit to which it may converge.

The fact (contained in Theorem 3.11) that a sequence converges in $\mathbb{R}^k$ if and only if it is a Cauchy sequence is usually called the *Cauchy criterion* for convergence.

Proof.

- (a) If $p_n \rightarrow p$ and if $\varepsilon > 0$, there is an integer N such that $d(p, p_n) < \varepsilon$ for all $n \geq N$. Hence

$$d(p_n, p_m) \leq d(p_n, p) + d(p, p_m) < 2\varepsilon$$

as soon as $n \geq N$ and $m \geq N$. Thus (p_n) is a Cauchy sequence.

- (b) Let (p_n) be a Cauchy sequence in the compact space X . For $N = 1, 2, 3, \dots$, let E_N be the set consisting of $p_N, p_{N+1}, p_{N+2}, \dots$. Then

$$\lim_{N \rightarrow \infty} \text{diam } \overline{E}_N = 0, \quad (3.3)$$

by Definition 3.9 and Theorem 3.10(a). Being a closed subset of a compact space X , each $\overline{E}_N$ is compact (Theorem 2.35). Also, $E_N \supset E_{N+1}$, so that $\overline{E}_N \supset \overline{E}_{N+1}$.

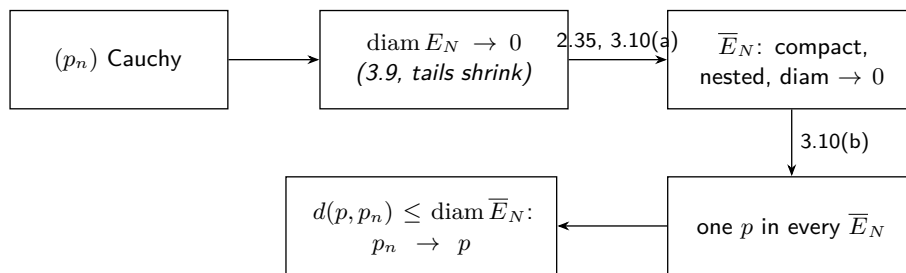
Theorem 3.10(b) shows now that there is a unique $p \in X$ which lies in every $\overline{E}_N$.¹⁶

Let $\varepsilon > 0$ be given. By (3.3) there is an integer N_0 such that $\text{diam } \overline{E}_N < \varepsilon$ if $N \geq N_0$. Since $p \in \overline{E}_N$, it follows that $d(p, q) < \varepsilon$ for every $q \in \overline{E}_N$, hence for every $q \in E_N$. In other words, $d(p, p_n) < \varepsilon$ if $n \geq N_0$. This says precisely that $p_n \rightarrow p$.

- (c) Let $(\mathbf{x}_n)$ be a Cauchy sequence in $\mathbb{R}^k$. Define E_N as in (b), with $\mathbf{x}_i$ in place of p_i . For some N , $\text{diam } E_N < 1$. The range of $(\mathbf{x}_n)$ is the union of E_N and the finite set $\{\mathbf{x}_1, \dots, \mathbf{x}_{N-1}\}$. Hence $(\mathbf{x}_n)$ is bounded. Since every bounded subset of $\mathbb{R}^k$ has compact closure in $\mathbb{R}^k$ (Theorem 2.41), (c) follows from (b).¹⁷

16 Why the closures? E_N itself need not be compact — it is just a scatter of points — and 3.10(b) needs compact sets. Passing to $\overline{E}_N$ costs nothing precisely because of 3.10(a): closing a set does not increase its diameter. That is the entire reason 3.10(a) was proved.

17 The structure of (c): Cauchy $\Rightarrow$ bounded (cheaply, from one E_N of finite diameter plus finitely many strays — the same split as 3.2(c)), then bounded $\Rightarrow$ inside a compact set (2.41), then apply (b). All the completeness is imported through 2.41.



Theorem 3.11(b) as a flow diagram: the tail-sets carry the whole argument, closures make them compact, and the nested-set theorem hands over the limit. Part (c) bolts “Cauchy $\Rightarrow$ bounded $\Rightarrow$ inside a compact set (2.41)” onto the front.

3.12 Definition. A metric space in which every Cauchy sequence converges is said to be *complete*.

Thus Theorem 3.11 says that all compact metric spaces and all euclidean spaces are complete. Theorem 3.11 implies also that every closed subset E of a complete metric space X is complete. (Every Cauchy sequence in E is a Cauchy sequence in X , hence it converges to some

$p \in X$, and actually $p \in E$ since E is closed.) An example of a metric space which is not complete is the space of all rational numbers, with $d(x, y) = |x - y|$.

The word “complete,” finally — and the full circle

This is the definition promised since Chapter 1. The comparison box at 1.10 said the least-upper-bound property “is what the word *complete* will name in 3.12”; here is the naming, and with it the last of the standard completeness statements enters the book. Of the ECNU text’s six equivalent formulations, five have now appeared: the supremum principle (1.19), nested intervals (2.38), Heine–Borel (2.41), Bolzano–Weierstrass (2.42 and 3.6(b)), and now the Cauchy criterion (3.11(c)); monotone convergence follows as 3.14. There is also a genuine full circle here. Tao *constructs* $\mathbb{R}$ as the completion of $\mathbb{Q}$ — real numbers are, in his book, equivalence classes of Cauchy sequences of rationals, so “every Cauchy sequence converges” is nearly true by construction. Rudin travelled the other way: $\mathbb{R}$ was characterised by suprema, and Cauchy completeness arrived at the end of a chain through compactness. Exercise 3.24 closes the loop from Rudin’s side, carrying out the Cauchy-sequence completion for an arbitrary metric space. One caution: completeness and compactness part company here. Compact implies complete (3.11(b)), but $\mathbb{R}^k$ is complete without being compact, and completeness is *not* intrinsic in the sense of 2.33 — $(0, 1)$ and $\mathbb{R}$ are homeomorphic, yet one is complete and the other not. Completeness belongs to the metric, not the topology.

Theorem 3.2(c) and example (d) after Definition 3.1 show that convergent sequences are bounded, but that bounded sequences in $\mathbb{R}^k$ need not converge. However, there is one important case in which convergence is equivalent to boundedness; this happens for monotonic sequences in $\mathbb{R}^1$.

3.13 Definition. A sequence (s_n) of real numbers is said to be

- (a) *monotonically increasing* if $s_n \leq s_{n+1}$ ($n = 1, 2, 3, \dots$);
- (b) *monotonically decreasing* if $s_n \geq s_{n+1}$ ($n = 1, 2, 3, \dots$).

The class of *monotonic* sequences consists of the increasing and the decreasing sequences.

3.14 Theorem. Suppose (s_n) is monotonic. Then (s_n) converges if and only if it is bounded.

Proof. Suppose $s_n \leq s_{n+1}$ (the proof is analogous in the other case). Let E be the range of (s_n) . If (s_n) is bounded, let s be the least upper bound of E . Then

$$s_n \leq s \quad (n = 1, 2, 3, \dots).$$

For every $\varepsilon > 0$, there is an integer N such that

$$s - \varepsilon < s_N \leq s,$$

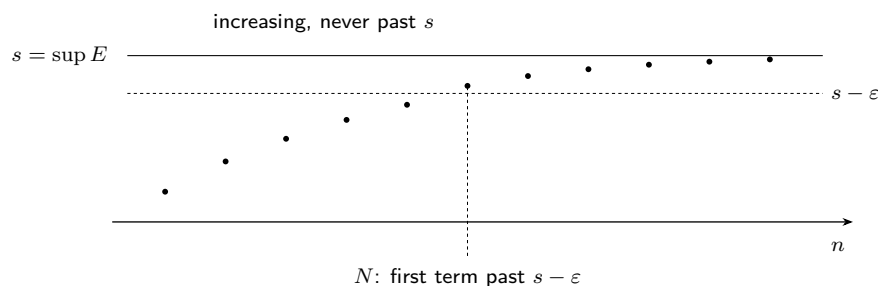
for otherwise $s - \varepsilon$ would be an upper bound of E .¹⁸ Since (s_n) increases, $n \geq N$ therefore implies

$$s - \varepsilon < s_n \leq s,$$

which shows that (s_n) converges (to s).

The converse follows from Theorem 3.2(c). $\square$

18 Clause (ii) of Definition 1.8 yet again, in its working form — step back from the supremum and some term sticks out past you. Monotonicity then does what no other hypothesis could: one term beyond $s - \varepsilon$ drags the entire tail with it.



Monotone convergence: the terms climb, the supremum caps them, and once one term passes $s - \varepsilon$ the rest are trapped in the gap. The limit is the supremum — 3.14 is Definition 1.10 restated for sequences, which is why the ECNU text lists the two as equivalent completeness axioms.

UPPER AND LOWER LIMITS

THE HARDEST DEFINITION IN THE CHAPTER, AND WHAT IT IS FOR

A divergent sequence can still have structure: $(-1)^n$ oscillates between two values rather than wandering. $\limsup$ and $\liminf$ capture the outer envelope of that oscillation — the largest and smallest subsequential limits. Their virtue is *unconditional existence*: every real sequence has both, in the extended reals, with no hypothesis at all. That is why the root test (3.33) and the radius-of-convergence formula (3.39) are stated with $\limsup$ — an ordinary limit might not exist, and the theory would fill up with exceptions. Keep Theorem 3.17 in view as the working characterisation: s^* is the one number that is both *reached* (along some subsequence) and *barely exceeded* (only finitely many terms above any $x > s^*$). The definition via $\sup E$ is for existence; 3.17 is what proofs actually use.

3.15 Definition. Let (s_n) be a sequence of real numbers with the following property: For every real M there is an integer N such that $n \geq N$ implies $s_n \geq M$. We then write

$$s_n \rightarrow +\infty.$$

Similarly, if for every real M there is an integer N such that $n \geq N$ implies $s_n \leq M$, we write

$$s_n \rightarrow -\infty.$$

It should be noted that we now use the symbol $\rightarrow$ (introduced in Definition 3.1) for certain types of divergent sequences, as well as for convergent sequences, but that the definitions of convergence and of limit, given in Definition 3.1, are unchanged.¹⁹

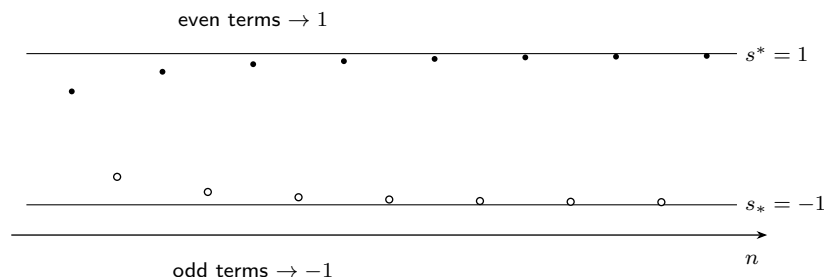
3.16 Definition. Let (s_n) be a sequence of real numbers. Let E be the set of numbers x (in the extended real number system) such that $s_{n_k} \rightarrow x$ for some subsequence (s_{n_k}) . This set E contains all subsequential limits as defined in Definition 3.5, plus possibly the numbers $+\infty$, $-\infty$.

We now recall Definitions 1.8 and 1.23 and put

$$s^* = \sup E, \quad s_* = \inf E.$$

The numbers s^* , s_* are called the *upper* and *lower limits* of (s_n) ; we use the notation

$$\limsup_{n \rightarrow \infty} s_n = s^*, \quad \liminf_{n \rightarrow \infty} s_n = s_*.$$



Example 3.18(b), $s_n = (-1)^n / (1 + 1/n)$: two subsequences, two subsequential limits, $E = \{-1, 1\}$. The upper and lower limits are the envelope the oscillation settles into — not the largest and smallest terms.

3.17 Theorem. Let (s_n) be a sequence of real numbers. Let E and s^* have the same meaning as in Definition 3.16. Then s^* has the following two properties:

- (a) $s^* \in E$.
 - (b) If $x > s^*$, there is an integer N such that $n \geq N$ implies $s_n < x$.
- Moreover, s^* is the only number with the properties (a) and (b).

Of course, an analogous result is true for s_* .

Proof.

- (a) If $s^* = +\infty$, then E is not bounded above; hence (s_n) is not bounded above, and there is a subsequence (s_{n_k}) such that $s_{n_k} \rightarrow +\infty$.

If s^* is real, then E is bounded above, and at least one subsequential limit exists, so that (a) follows from Theorems 3.7 and 2.28.²⁰

19 So “ $s_n \rightarrow +\infty$ ” and “ (s_n) diverges” are compatible statements — the arrow has been extended, the word limit has not. This is the extended-real bookkeeping of 1.23 paying its way: $+\infty$ is a destination for notation, not a number a sequence can converge to.

20 Here is where the section’s stockpile is spent, both pieces at once: 3.7 says E is closed, 2.28 says $\sup E$ lies in $\overline{E} = E$. So the supremum of the subsequential limits is itself a subsequential limit — attained, not merely approached. This is the fact that makes $\limsup$ usable.

If $s^* = -\infty$, then E contains only one element, namely $-\infty$, and there is no subsequential limit. Hence, for any real M , $s_n > M$ for at most a finite number of values of n , so that $s_n \rightarrow -\infty$.

This establishes (a) in all cases.

- (b) Suppose there is a number $x > s^*$ such that $s_n \geq x$ for infinitely many values of n . In that case, there is a number $y \in E$ such that $y \geq x > s^*$, contradicting the definition of s^* .²¹

Thus s^* satisfies (a) and (b). To show the uniqueness, suppose there are two numbers, p and q , which satisfy (a) and (b), and suppose $p < q$. Choose x such that $p < x < q$. Since p satisfies (b), we have $s_n < x$ for $n \geq N$. But then q cannot satisfy (a). $\square$

THREE WORKING FORMS OF ONE NUMBER

$\limsup$ wears three interchangeable outfits, and fluency means switching freely.

As $\sup E$ (Definition 3.16): best for existence and for symmetry with $\liminf$.

As the two-sided test (Theorem 3.17): s^* is reached along a subsequence, and any $x > s^*$ eventually caps the whole sequence. In ε -form: for every $\varepsilon > 0$, $s_n > s^* + \varepsilon$ only finitely often, while $s_n > s^* - \varepsilon$ infinitely often. This is the form the root test uses.

As a limit of suprema: $s^* = \lim_{N \rightarrow \infty} \sup_{n \geq N} s_n$ — the tail-suprema $\sup_{n \geq N} s_n$ decrease as the tail shrinks, so they converge by 3.14, and their limit is s^* . Rudin leaves this form to Exercise 3.5's neighbourhood; Tao takes it as the *definition* (his 6.4.6, via “sup of the tail”), and the ECNU text develops it in its §7.2. It explains the name: the *lim* of the *sup*s.

21 The compressed step: infinitely many terms $\geq x$ form a subsequence in $[x, +\infty]$, and 3.6/3.17(a) applied to it yields a subsequential limit $y \geq x$. “Infinitely many terms above x ” always hands you an element of E at or above x — that conversion, bounded-infinite-to-limit, is Bolzano–Weierstrass 3.6(b) on duty.

3.18 Examples.

- (a) Let (s_n) be a sequence containing all rationals. Then every real number is a subsequential limit, and

$$\limsup_{n \rightarrow \infty} s_n = +\infty, \quad \liminf_{n \rightarrow \infty} s_n = -\infty.$$

22

- (b) Let $s_n = (-1)^n / [1 + (1/n)]$. Then

$$\limsup_{n \rightarrow \infty} s_n = 1, \quad \liminf_{n \rightarrow \infty} s_n = -1.$$

- (c) For a real-valued sequence (s_n) , $\lim_{n \rightarrow \infty} s_n = s$ if and only if

$$\limsup_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_n = s.$$

23

22 Such a sequence exists because $\mathbb{Q}$ is countable (Corollary to 2.13), and every real is a subsequential limit because $\mathbb{Q}$ is dense (1.20(b)) — the two headline facts about $\mathbb{Q}$, cooperating. Here E is all of the extended line: divergence as total as it gets.

23 The squeeze that makes the pair useful: convergence is exactly “the envelope closes.” Many later proofs show $\limsup \leq c$ and $\liminf \geq c$ separately and conclude $\lim = c$; 3.19 exists to service such arguments.

We close this section with a theorem which is useful and whose proof is quite trivial:

3.19 Theorem. *If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then*

$$\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n, \quad \limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n.$$

SOME SPECIAL SEQUENCES

A TOOLKIT, ASSEMBLED BY ONE METHOD | Five limits used from here to the end of the book, each proved by the squeeze stated in the opening line: trap x_n between 0 and something already known to vanish. The recurring supplier of the trap is the binomial theorem — expand $(1+x)^n$, keep a single term, and discard the rest. Item (d) is the one to remember: powers beat polynomials, and (e), geometric decay, is its special case $\alpha = 0$. These feed directly into 3.26 and every comparison with a geometric series afterwards.

We shall now compute the limits of some sequences which occur frequently. The proofs will all be based on the following remark: *If $0 \leq x_n \leq s_n$ for $n \geq N$, where N is some fixed number, and if $s_n \rightarrow 0$, then $x_n \rightarrow 0$.*

3.20 Theorem.

- (a) *If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.*
- (b) *If $p > 0$, then $\lim_{n \rightarrow \infty} \sqrt[p]{p} = 1$.*
- (c) $\lim_{n \rightarrow \infty} \sqrt[p]{n} = 1$.
- (d) *If $p > 0$ and α is real, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$.*
- (e) *If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$.*

Proof.

- (a) Take $n > (1/\varepsilon)^{1/p}$. (Note that the archimedean property of the real number system is used here.)²⁴
- (b) If $p > 1$, put $x_n = \sqrt[p]{p} - 1$. Then $x_n > 0$ and, by the binomial theorem,

$$1 + nx_n \leq (1 + x_n)^n = p,$$

so that

$$0 < x_n \leq \frac{p-1}{n}.$$

Hence $x_n \rightarrow 0$. If $p = 1$, (b) is trivial, and if $0 < p < 1$, the result is obtained by taking reciprocals.

²⁴ And note what else is used: n^p for real p is Exercise 1.6's construction, and the inequality manipulation needs 1.21. The one-line proof stands on the chapter-1 exercises.

(c) Put $x_n = \sqrt[n]{n} - 1$. Then $x_n \geq 0$ and, by the binomial theorem,

$$n = (1 + x_n)^n \geq \frac{n(n-1)}{2} x_n^2.$$

Hence

$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}} \quad (n \geq 2).$$

25

(d) Let k be an integer such that $k > \alpha$, $k > 0$. For $n > 2k$,

$$(1+p)^n > \binom{n}{k} p^k = \frac{n(n-1) \cdots (n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}.$$

Hence

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k} \quad (n > 2k).$$

Since $\alpha - k < 0$, $n^{\alpha-k} \rightarrow 0$, by (a).

(e) Take $\alpha = 0$ in (d).

□

25 Same trick as (b), one term deeper. Keeping the linear term gives only $x_n \leq (n-1)/n$, which is useless; the quadratic term grows like n^2 and beats the left side. Choosing which binomial term to keep is the entire skill, and (d) makes the choice a third time, with term k .

SERIES

A SERIES IS A SEQUENCE IN DISGUISE | Nothing new is defined in this section except notation: a series *is* its sequence of partial sums, and Rudin says so twice — in 3.21 (“ s is the limit of a sequence of sums, and is not obtained simply by addition”) and in the remark that every theorem about sequences can be restated for series via $a_n = s_n - s_{n-1}$. The three theorems here are the three big sequence facts passed through that dictionary: 3.22 is the Cauchy criterion 3.11, 3.24 is monotone convergence 3.14, and 3.23 is the trivial-but-vital corollary that terms of a convergent series must die out.

The section ends with the comparison test, 3.25 — the master test. Every test that follows (root, ratio, condensation applications) works by comparing against a series already understood, almost always the geometric series 3.26.

In the remainder of this chapter, all sequences and series under consideration will be complex-valued, unless the contrary is explicitly stated. Extensions of some of the theorems which follow, to series with terms in $\mathbb{R}^k$, are mentioned in Exercise 3.15.

3.21 Definition. Given a sequence (a_n) , we use the notation

$$\sum_{n=p}^q a_n \quad (p \leq q)$$

to denote the sum $a_p + a_{p+1} + \cdots + a_q$. With (a_n) , we associate a sequence (s_n) , where

$$s_n = \sum_{k=1}^n a_k.$$

For (s_n) we also use the symbolic expression

$$a_1 + a_2 + a_3 + \cdots$$

or, more concisely,

$$\sum_{n=1}^{\infty} a_n. \quad (3.4)$$

The symbol in (3.4) we call an *infinite series*, or just a *series*. The numbers s_n are called the *partial sums* of the series. If (s_n) converges to s , we say that the series *converges*, and write

$$\sum_{n=1}^{\infty} a_n = s.$$

The number s is called the *sum* of the series; but it should be clearly understood that s is the limit of a sequence of sums, and is not obtained simply by addition.²⁶

If (s_n) diverges, the series is said to *diverge*.

Sometimes, for convenience of notation, we shall consider series of the form

$$\sum_{n=0}^{\infty} a_n. \quad (3.5)$$

And frequently, when there is no possible ambiguity, or when the distinction is immaterial, we shall simply write Σa_n in place of (3.4) or (3.5).

It is clear that every theorem about sequences can be stated in terms of series (putting $a_1 = s_1$ and $a_n = s_n - s_{n-1}$ for $n > 1$), and vice versa. But it is nevertheless useful to consider both concepts.

The Cauchy criterion (Theorem 3.11) can be restated in the following form:

3.22 Theorem. Σa_n converges if and only if for every $\varepsilon > 0$ there is an integer N such that

$$\left| \sum_{k=n}^m a_k \right| \leq \varepsilon \quad (3.6)$$

if $m \geq n \geq N$.

In particular, by taking $m = n$, (3.6) becomes

$$|a_n| \leq \varepsilon \quad (n \geq N).$$

In other words:

26 This sentence is the guard rail for the whole second half of the chapter. Addition of finitely many numbers is commutative and associative by the field axioms; a “sum” that is really a limit inherits neither property automatically. Theorem 3.54 will show rearrangement genuinely failing, and 3.55 will show what hypothesis buys it back.

3.23 Theorem. *If Σa_n converges, then $\lim_{n \rightarrow \infty} a_n = 0$.*

The condition $a_n \rightarrow 0$ is not, however, sufficient to ensure convergence of Σa_n . For instance, the series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges; for the proof, we refer to Theorem 3.28.²⁷

Theorem 3.14, concerning monotonic sequences, also has an immediate counterpart for series.

3.24 Theorem. *A series of nonnegative¹ terms converges if and only if its partial sums form a bounded sequence.*²⁸

We now turn to a convergence test of a different nature, the so-called “comparison test.”

3.25 Theorem.

(a) *If $|a_n| \leq c_n$ for $n \geq N_0$, where N_0 is some fixed integer, and if Σc_n converges, then Σa_n converges.*

(b) *If $a_n \geq d_n \geq 0$ for $n \geq N_0$ and if Σd_n diverges, then Σa_n diverges.*

Note that (b) applies only to series of nonnegative terms a_n .

Proof. Given $\varepsilon > 0$, there exists $N \geq N_0$ such that $m \geq n \geq N$ implies

$$\sum_{k=n}^m c_k \leq \varepsilon,$$

by the Cauchy criterion. Hence

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k| \leq \sum_{k=n}^m c_k \leq \varepsilon,$$

and (a) follows.

Next, (b) follows from (a), for if Σa_n converges, so must Σd_n [note that (b) also follows from Theorem 3.24]. $\square$

WHAT COMPARISON ACTUALLY PROVES, AND THE ASYMMETRY

Look at the chain of inequalities: it bounds $|\sum a_k|$ through $\sum |a_k|$. So (a) silently proves more than it claims — under its hypotheses $\Sigma |a_n|$ converges too, i.e. the convergence is *absolute* (3.45’s terminology). This is why Rudin will say at 3.46 that comparison, root and ratio “cannot give any information about conditionally convergent series”: all three run through this chain.

The asymmetry between (a) and (b) is not decoration. Domination transfers convergence *downward* and divergence *upward*, never the reverse; and (b) needs $a_n \geq 0$ because a wildly oscillating series can

27 *Necessary, not sufficient — the single most-cited warning in elementary analysis. The harmonic series is the canonical witness: terms shrink to 0, partial sums climb past every bound. Rudin flags it before anyone can form the converse habit; 3.22 with $m > n$ is what genuine convergence requires, and a block of small terms can still add up.*

28 *Nonnegative terms make the partial sums monotonically increasing, and 3.14 does the rest. This innocent-looking equivalence is the engine of the whole nonnegative theory: comparison (3.25), condensation (3.27), and the root test all reduce convergence to boundedness of partial sums, which is checked by comparing.*

¹The expression “nonnegative” always refers to real numbers.

dominate a divergent positive one and still converge — cancellations rescue it. Keep the mantra: below a convergent series, convergent; above a divergent nonnegative series, divergent.

The comparison test is a very useful one; to use it efficiently, we have to become familiar with a number of series of nonnegative terms whose convergence or divergence is known.

SERIES OF NONNEGATIVE TERMS

The simplest of all is perhaps the geometric series.

3.26 Theorem. *If $0 \leq x < 1$, then*

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

If $x \geq 1$, the series diverges.

Proof. If $x \neq 1$,

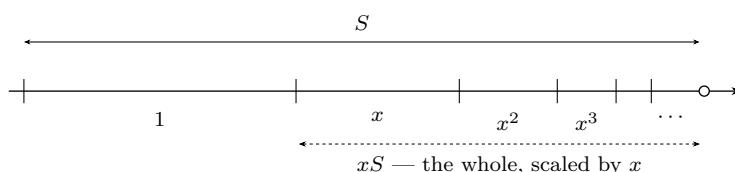
$$s_n = \sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}.$$

The result follows if we let $n \rightarrow \infty$. For $x = 1$, we get

$$1 + 1 + 1 + \cdots,$$

which evidently diverges.²⁹ □

²⁹ One of only two series in the chapter whose sum is computed rather than merely shown to exist (the other is e). The closed form for s_n — multiply by x , subtract, telescope — is the entire proof, and the limit uses 3.20(e). Every later test converges things by comparing them to this one series; it is the yardstick.



The segments $1, x, x^2, \dots$ laid end to end (drawn for $x = 0.6$). The tail beyond the first segment is a copy of the entire picture shrunk by the factor x , so $S = 1 + xS$, i.e. $S = 1/(1 - x)$. Rudin's closed form for s_n is this picture stopped after n steps — it is the identity of 1.21, $b^{n+1} - a^{n+1} = (b - a)(b^n + \cdots + a^n)$, with $b = 1$, $a = x$.

In many cases which occur in applications, the terms of the series decrease monotonically. The following theorem of Cauchy is therefore of particular interest. The striking feature of the theorem is that a rather “thin” subsequence of (a_n) determines the convergence or divergence of $\sum a_n$.

3.27 Theorem. *Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series*

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots \quad (3.7)$$

converges.

Proof. By Theorem 3.24, it suffices to consider the boundedness of the partial sums. Let

$$\begin{aligned}s_n &= a_1 + a_2 + \cdots + a_n, \\ t_k &= a_1 + 2a_2 + \cdots + 2^k a_{2^k}.\end{aligned}$$

For $n < 2^k$,

$$\begin{aligned}s_n &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + \cdots + 2^k a_{2^k} \\ &= t_k,\end{aligned}$$

so that

$$s_n \leq t_k. \quad (3.8)$$

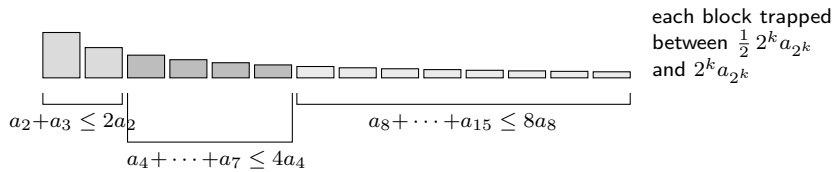
On the other hand, if $n > 2^k$,

$$\begin{aligned}s_n &\geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + \cdots + 2^{k-1}a_{2^k} \\ &= \frac{1}{2}t_k,\end{aligned}$$

so that

$$2s_n \geq t_k. \quad (3.9)$$

By (3.8) and (3.9), the sequences (s_n) and (t_k) are either both bounded or both unbounded. This completes the proof. $\square$



Cauchy condensation: chop the series into blocks of doubling length. Monotonicity makes each block at most its first term times its length, and at least its last term times half its length — inequalities (3.8) and (3.9).

Why the sampling is allowed to be so thin

The theorem replaces all of (a_n) by the sample $a_1, a_2, a_4, a_8, \dots$ — vanishingly sparse — and loses nothing. Monotonicity is what pays for it: within a dyadic block the terms cannot vary by more than the endpoints, so the whole block is measured by one sample to within a factor of 2. Without monotonicity the sampled series says nothing (put all the mass off the sample points).

The choice of base 2 is a convenience; any factor works. And the theorem is built for exactly one purpose, visible in 3.28–3.29: it turns $\sum 1/n^p$ into a *geometric* series, and $\sum 1/[n(\log n)^p]$ into $\sum 1/k^p$ —

each application of condensation strips off one logarithm. That is why the borderline series in Rudin's ladder all involve iterated logs.

3.28 Theorem. $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Proof. If $p \leq 0$, divergence follows from Theorem 3.23. If $p > 0$, Theorem 3.27 is applicable, and we are led to the series

$$\sum_{k=0}^{\infty} 2^k \cdot \frac{1}{2^{kp}} = \sum_{k=0}^{\infty} 2^{(1-p)k}.$$

Now, $2^{1-p} < 1$ if and only if $1 - p < 0$, and the result follows by comparison with the geometric series (take $x = 2^{1-p}$ in Theorem 3.26).³⁰

□

As a further application of Theorem 3.27, we prove:

3.29 Theorem. If $p > 1$,

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \quad (3.10)$$

converges; if $p \leq 1$, the series diverges.

Remark: “ $\log n$ ” denotes the logarithm of n to the base e (compare Exercise 1.7); the number e will be defined in a moment (see Definition 3.30). We let the series start with $n = 2$, since $\log 1 = 0$.

Proof. The monotonicity of the logarithmic function (which will be discussed in more detail in Chap. 8) implies that $(\log n)_{n=1}^{\infty}$ increases. Hence $(1/[n(\log n)^p])_{n=1}^{\infty}$ decreases, and we can apply Theorem 3.27 to (3.10); this leads us to the series

$$\sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k (\log 2^k)^p} = \sum_{k=1}^{\infty} \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}, \quad (3.11)$$

and Theorem 3.29 follows from Theorem 3.28. □

This procedure may evidently be continued. For instance,

$$\sum_{n=3}^{\infty} \frac{1}{n \log n \log(\log n)} \quad (3.12)$$

diverges, whereas

$$\sum_{n=3}^{\infty} \frac{1}{n \log n (\log(\log n))^2} \quad (3.13)$$

converges.

We may now observe that the terms of the series (3.12) differ very little from those of (3.13). Still, one diverges, the other converges. If we continue the process which led us from Theorem 3.28 to Theorem 3.29, and then to (3.12) and (3.13), we get pairs of convergent and

³⁰ With $p = 1$ this settles the harmonic series: condensation turns it into $\sum_k 2^k \cdot 2^{-k} = 1 + 1 + 1 + \cdots$. The classical grouped proof — $\frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \cdots + \frac{1}{8}) + \cdots \geq \frac{1}{2} + \frac{1}{2} + \cdots$ — is exactly inequality (3.9) written out for this one series.

divergent series whose terms differ even less than those of (3.12) and (3.13). One might thus be led to the conjecture that there is a limiting situation of some sort, a “boundary” with all convergent series on one side, all divergent series on the other side — at least as far as series with monotonic coefficients are concerned. This notion of a “boundary” is of course quite vague. The point we wish to make is this: *No matter how we make this notion precise, the conjecture is false.* Exercises 3.11(b) and 3.12(b) may serve as illustrations.³¹

We do not wish to go any deeper into this aspect of convergence theory, and refer the reader to Knopp’s “Theory and Application of Infinite Series,” Chap. IX, particularly Sec. 41.

THE NUMBER e

WHY e APPEARS IN A CHAPTER ON SERIES | Partly because it can — the series $\sum 1/n!$ is the book’s first named sum — and partly as a demonstration piece. Theorem 3.31 reconciles the two classical definitions of e using exactly the $\limsup/\liminf$ machinery just built, and Rudin flags it as “a good illustration of operations with limits.” The error estimate (3.16) then delivers the irrationality of e in five lines, the chapter’s most elegant single proof. Yu Pin’s Tsinghua course takes the definition of e (and eventually $e^{i\pi} = -1$) as the organising goal of its entire first semester; Rudin gets the number itself this cheaply because the series technology came first.

3.30 Definition. $e = \sum_{n=0}^{\infty} \frac{1}{n!}$.

Here $n! = 1 \cdot 2 \cdot 3 \cdots n$ if $n \geq 1$, and $0! = 1$. Since

$$\begin{aligned} s_n &= 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} < 3, \end{aligned}$$

the series converges, and the definition makes sense.³² In fact, the series converges very rapidly and allows us to compute e with great accuracy.

It is of interest to note that e can also be defined by means of another limit process; the proof provides a good illustration of operations with limits:

3.31 Theorem. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let

$$s_n = \sum_{k=0}^n \frac{1}{k!}, \quad t_n = \left(1 + \frac{1}{n}\right)^n.$$

[31] A rare editorial aside from Rudin, and worth taking seriously: there is no slowest divergent series and no fastest convergent one. Given any convergent $\sum a_n$ with positive terms, Exercise 3.12(b) produces a convergent series with terms larger by an unbounded factor; 3.11(b) does the dual for divergence. The convergence/divergence frontier is not a curve one can creep up to — a genuinely counterintuitive fact, and the reason “comparison with the worst convergent series” can never be a test.

32 The comparison is with the geometric series, of course: $n! \geq 2^{n-1}$ because each factor past the first is at least 2. Convergence via 3.24 — increasing partial sums, bounded by 3. So $2 < e < 3$ before anything else is known about it.

By the binomial theorem,

$$t_n = 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \cdots \\ + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right).$$

Hence $t_n \leq s_n$, so that

$$\limsup_{n \rightarrow \infty} t_n \leq e, \quad (3.14)$$

by Theorem 3.19. Next, if $n \geq m$,

$$t_n \geq 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!} \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right).$$

Let $n \rightarrow \infty$, keeping m fixed. We get

$$\liminf_{n \rightarrow \infty} t_n \geq 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{m!},$$

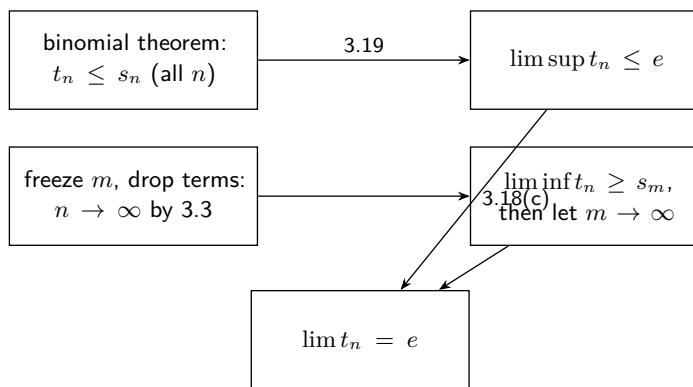
so that

$$s_m \leq \liminf_{n \rightarrow \infty} t_n.$$

Letting $m \rightarrow \infty$, we finally get

$$e \leq \liminf_{n \rightarrow \infty} t_n. \quad (3.15)$$

The theorem follows from (3.14) and (3.15). $\square$



Two independent one-sided bounds, closed by the squeeze of 3.18(c). The upper branch may use all n at once; the lower must freeze m first — the legal order for exchanging two limits.

THE TWO-PARAMETER SQUEEZE, DISSECTED

The proof is a small masterclass in a manoeuvre used constantly from here on. The obstacle: t_n is a sum of n terms, each *changing with n* , so no single limit theorem applies. The solution: cut the dependence in two.

Upper half. Every bracketed factor is ≤ 1 , so term by term $t_n \leq$

s_n ; take $\limsup$ (which exists unconditionally — this is why 3.16 was built) and use 3.19.

Lower half. Freeze the number of terms at m , discard the rest — legitimate since all terms are positive — and let only n run. Now there are finitely many factors, each $\rightarrow 1$, so 3.3 applies and gives $\liminf t_n \geq s_m$ for each *fixed* m . Only then release m .

The order of operations is everything: n first with m frozen, m second. Interchanging two limit processes illegally is the classic analysis error, and this proof is the book's first demonstration of how to do it legally — by one-sided bounds and a squeeze. Example 3.18(c) closes the trap: $\limsup = \liminf = e$ forces $\lim = e$.

The rapidity with which the series $\sum 1/n!$ converges can be estimated as follows: If s_n has the same meaning as above, we have

$$\begin{aligned} e - s_n &= \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots \\ &< \frac{1}{(n+1)!} \left\{ 1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots \right\} = \frac{1}{n!n} \end{aligned}$$

so that

$$0 < e - s_n < \frac{1}{n!n}. \quad (3.16)$$

Thus s_{10} , for instance, approximates e with an error less than 10^{-7} . The inequality (3.16) is of theoretical interest as well, since it enables us to prove the irrationality of e very easily.

3.32 Theorem. e is irrational.

Proof. Suppose e is rational. Then $e = p/q$, where p and q are positive integers. By (3.16),

$$0 < q!(e - s_q) < \frac{1}{q}. \quad (3.17)$$

By our assumption, $q!e$ is an integer. Since

$$q!s_q = q! \left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!} \right)$$

is an integer, we see that $q!(e - s_q)$ is an integer.

Since $q \geq 1$, (3.17) implies the existence of an integer between 0 and 1. We have thus reached a contradiction. $\square$

Actually, e is not even an algebraic number. For a simple proof of this, see page 25 of Niven's book, or page 176 of Herstein's, cited in the Bibliography.

The shape of the irrationality proof

Compare Chapter 1's proof that $\sqrt{2} \notin \mathbb{Q}$: there, rationality forced an infinite descent; here it forces an integer strictly between 0 and 1. Both are impossibility-of-an-integer arguments, and the second

pattern — scale the error by exactly the denominator’s factorial and watch it land in $(0, 1)$ — is the template for most irrationality proofs in analysis.

Everything hinges on *how fast* the series converges: (3.16) says the tail after s_q is smaller than $1/(q!q)$, so multiplying by $q!$ leaves something below $1/q$, still positive. A series converging slower — like $\sum 1/n^2$, whose sum $\pi^2/6$ is also irrational — gives no such one-line proof. Fast convergence is arithmetic leverage: Exercise 2.3 showed transcendental numbers *exist* without naming one, and here analysis pins down an explicit number as irrational.

THE ROOT AND RATIO TESTS

TWO COMPARISONS WITH THE GEOMETRIC SERIES | Neither test is a new idea. Both ask one question — does Σa_n sit below some geometric series $\Sigma \beta^n$ with $\beta < 1$? — and differ only in how they detect it: the root test reads the geometric rate off $\sqrt[n]{|a_n|}$ directly, the ratio test off the quotient of consecutive terms. This is why $\limsup$ was built: the rates need not converge, and 3.17(b) is exactly the tool that converts “ $\limsup < 1$ ” into “eventually below a fixed $\beta < 1$.”

The relationship between the two is settled by 3.37: the root test is strictly stronger, and Example 3.35 exhibits a series the ratio test cannot touch. Use the ratio test for factorials and products, the root test for theory — it is the one the radius of convergence (3.39) is built on.

3.33 Theorem. Given Σa_n , put $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$.

- (a) If $\alpha < 1$, Σa_n converges;
- (b) if $\alpha > 1$, Σa_n diverges;
- (c) if $\alpha = 1$, the test gives no information.

Proof. If $\alpha < 1$, we can choose β so that $\alpha < \beta < 1$, and an integer N such that

$$\sqrt[n]{|a_n|} < \beta$$

for $n \geq N$ [by Theorem 3.17(b)]. That is, $n \geq N$ implies

$$|a_n| < \beta^n.$$

Since $0 < \beta < 1$, $\Sigma \beta^n$ converges. Convergence of Σa_n follows now from the comparison test.³³

If $\alpha > 1$, then, again by Theorem 3.17, there is a sequence (n_k) such that

$$\sqrt[n_k]{|a_{n_k}|} \rightarrow \alpha.$$

33 Watch 3.17(b) do precisely its promised job: $\beta > \alpha = s^*$ caps the whole tail. The room between α and 1 is essential — the comparison is with $\Sigma \beta^n$ for a fixed $\beta < 1$, which is why $\alpha = 1$ proves nothing: no room remains.

Hence $|a_n| > 1$ for infinitely many values of n , so that the condition $a_n \rightarrow 0$, necessary for convergence of $\sum a_n$, does not hold (Theorem 3.23).

To prove (c), we consider the series

$$\sum \frac{1}{n}, \quad \sum \frac{1}{n^2}.$$

For each of these series $\alpha = 1$, but the first diverges and the second converges.³⁴ $\square$

3.34 Theorem. *The series $\sum a_n$*

- (a) *converges if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$,*
- (b) *diverges if $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq n_0$, where n_0 is some fixed integer.*

Proof. If condition (a) holds, we can find $\beta < 1$, and an integer N , such that

$$\left| \frac{a_{n+1}}{a_n} \right| < \beta$$

for $n \geq N$. In particular,

$$\begin{aligned} |a_{N+1}| &< \beta |a_N|, \\ |a_{N+2}| &< \beta |a_{N+1}| < \beta^2 |a_N|, \\ &\vdots \\ |a_{N+p}| &< \beta^p |a_N|. \end{aligned}$$

That is,

$$|a_n| < |a_N| \beta^{-N} \cdot \beta^n$$

for $n \geq N$, and (a) follows from the comparison test, since $\sum \beta^n$ converges.

If $|a_{n+1}| \geq |a_n|$ for $n \geq n_0$, it is easily seen that the condition $a_n \rightarrow 0$ does not hold, and (b) follows. $\square$

Note: the knowledge that $\lim a_{n+1}/a_n = 1$ implies nothing about the convergence of $\sum a_n$. The series $\sum 1/n$ and $\sum 1/n^2$ demonstrate this.

THE DIVERGENCE HALF OF THE RATIO TEST IS NOT A LIMSUP

Read (b) again: it demands $|a_{n+1}/a_n| \geq 1$ for all $n \geq n_0$ — a pointwise, eventually-everywhere condition. The tempting statement “ $\limsup |a_{n+1}/a_n| > 1$ implies divergence” is **false**, and Example 3.35(b) below is the counterexample: a convergent series with ratio $\limsup$ equal to 2. A few large ratios do no harm if they are followed by crashes; only a ratio that *never* again dips below 1 pins the terms away from zero.

Symmetrically, the convergence half really is a $\limsup$, and the $\liminf$ -

34 Both α 's are 1 by 3.20(c): $\sqrt[n]{1/n} = 1/\sqrt[n]{n} \rightarrow 1$, and likewise for $1/n^2$. The p -series live exactly on the geometric tests' blind spot — their terms decay polynomially, slower than every geometric rate, which is why 3.27/3.28 needed a different idea.

version “ $\liminf |a_{n+1}/a_n| < 1$ implies convergence” is false too — 3.35(a) has ratio $\liminf 0$ and could easily be built divergent. The test is asymmetric because it detects geometric *domination*, and domination is one-sided.

3.35 Examples.

(a) Consider the series

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots,$$

for which

$$\begin{aligned}\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0, \\ \liminf_{n \rightarrow \infty} \sqrt[n]{a_n} &= \lim_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{3^n}} = \frac{1}{\sqrt{3}}, \\ \limsup_{n \rightarrow \infty} \sqrt[n]{a_n} &= \lim_{n \rightarrow \infty} \sqrt[2n]{\frac{1}{2^n}} = \frac{1}{\sqrt{2}}, \\ \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} &= \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{3}{2}\right)^n = +\infty.\end{aligned}$$

35 The series interleaves two geometric series with different rates. The roots feel only the rates, and both are safely below 1; the ratios feel the jumps between the two strands, which grow without bound. Interleaving is the standard way to break the ratio test.

The root test indicates convergence; the ratio test does not apply.³⁵

(b) The same is true for the series

$$\frac{1}{2} + 1 + \frac{1}{8} + \frac{1}{4} + \frac{1}{32} + \frac{1}{16} + \frac{1}{128} + \frac{1}{64} + \cdots,$$

where

$$\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{8}, \quad \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 2,$$

but

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \frac{1}{2}.$$

3.36 Remarks. The ratio test is frequently easier to apply than the root test, since it is usually easier to compute ratios than n th roots. However, the root test has wider scope. More precisely: Whenever the ratio test shows convergence, the root test does too; whenever the root test is inconclusive, the ratio test is too. This is a consequence of Theorem 3.37 and is illustrated by the above examples.

Neither of the two tests is subtle with regard to divergence. Both deduce divergence from the fact that a_n does not tend to zero as $n \rightarrow \infty$.

3.37 Theorem. For any sequence (c_n) of positive numbers,

$$\begin{aligned}\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} &\leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n}, \\ \limsup_{n \rightarrow \infty} \sqrt[n]{c_n} &\leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.\end{aligned}$$

Proof. We shall prove the second inequality; the proof of the first is quite similar. Put

$$\alpha = \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.$$

If $\alpha = +\infty$, there is nothing to prove. If α is finite, choose $\beta > \alpha$. There is an integer N such that

$$\frac{c_{n+1}}{c_n} \leq \beta$$

for $n \geq N$. In particular, for any $p > 0$,

$$c_{N+k+1} \leq \beta c_{N+k} \quad (k = 0, 1, \dots, p-1).$$

Multiplying these inequalities, we obtain

$$c_{N+p} \leq \beta^p c_N,$$

or

$$c_n \leq c_N \beta^{-N} \cdot \beta^n \quad (n \geq N).$$

Hence

$$\sqrt[n]{c_n} \leq \sqrt[n]{c_N \beta^{-N}} \cdot \beta,$$

so that

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \beta, \quad (3.18)$$

by Theorem 3.20(b). Since (3.18) is true for every $\beta > \alpha$, we have

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \alpha.$$

36

□

POWER SERIES

SIX CHAPTERS OF CONSEQUENCES BEGIN HERE | Two items only — a definition and one theorem — but 3.39 is the foundation on which Chapter 8 builds the exponential, the trigonometric functions and eventually $e^{i\pi} = -1$. The theorem says the convergence domain of $\sum c_n z^n$ is always a disc, and computes its radius by the root test. That a single formula answers “for which z ?” is the first sign of how rigid power series are; the second, uniform convergence inside the disc, arrives at 8.1.

36 The mechanism: a bound on every ratio compounds into a geometric bound on the terms themselves, and taking n th roots flattens the constant $c_N \beta^{-N}$ into 1 by 3.20(b). Roots retain only the exponential rate; ratios carry every local fluctuation. That is the entire content of “root is stronger than ratio,” and it is also why $\lim c_{n+1}/c_n = L$ forces $\lim \sqrt[n]{c_n} = L$ — squeeze both displayed inequalities. Setting $c_n = n!/n^n$ gives the useful byproduct $\sqrt[n]{n!} \sim n/e$.

3.38 Definition. Given a sequence (c_n) of complex numbers, the series

$$\sum_{n=0}^{\infty} c_n z^n \quad (3.19)$$

is called a *power series*. The numbers c_n are called the *coefficients* of the series; z is a complex number.

In general, the series will converge or diverge, depending on the choice of z . More specifically, with every power series there is associated a circle, the *circle of convergence*, such that (3.19) converges if z is in the interior of the circle and diverges if z is in the exterior of the circle (to cover all cases, we have to consider the plane as the interior of a circle with infinite radius, and a point as a circle with radius zero). The behavior on the circle of convergence is much more varied and cannot be described so simply.

3.39 Theorem. Given the power series $\sum c_n z^n$, put

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}, \quad R = \frac{1}{\alpha}.$$

(If $\alpha = 0$, $R = +\infty$; if $\alpha = +\infty$, $R = 0$.) Then $\sum c_n z^n$ converges if $|z| < R$ and diverges if $|z| > R$.

Proof. Put $a_n = c_n z^n$ and apply the root test:

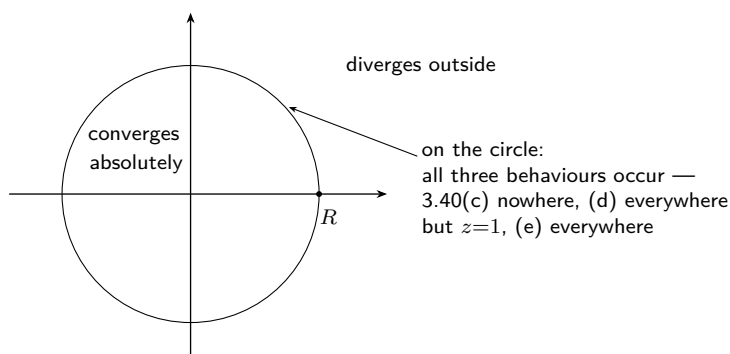
$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = |z| \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}.$$

□

Note: R is called the *radius of convergence* of $\sum c_n z^n$.

3.40 Examples.

- (a) The series $\sum n^n z^n$ has $R = 0$.
- (b) The series $\sum \frac{z^n}{n!}$ has $R = +\infty$. (In this case the ratio test is easier to apply than the root test.)
- (c) The series $\sum z^n$ has $R = 1$. If $|z| = 1$, the series diverges, since (z^n) does not tend to 0 as $n \rightarrow \infty$.
- (d) The series $\sum \frac{z^n}{n}$ has $R = 1$. It diverges if $z = 1$. It converges for all other z with $|z| = 1$. (The last assertion will be proved in Theorem 3.44.)
- (e) The series $\sum \frac{z^n}{n^2}$ has $R = 1$. It converges for all z with $|z| = 1$, by the comparison test, since $|z^n/n^2| = 1/n^2$.



Theorem 3.39: inside the circle everything converges (absolutely — the proof runs through the root test, hence 3.25), outside everything diverges, and the circle itself carries no general rule.

SUMMATION BY PARTS

THE ONE TOOL FOR CONDITIONAL CONVERGENCE | Rudin will observe at 3.46 that comparison, root and ratio only ever detect *absolute* convergence. This section supplies the single technique in the book that goes further. The identity 3.41 trades $\sum a_n b_n$ for a sum involving the partial sums A_n — profitable exactly when the A_n are bounded though $\sum a_n$ itself diverges, as with $a_n = (-1)^n$ or $a_n = z^n$ on the unit circle. Theorem 3.42 packages the trade; 3.43 (alternating series) and 3.44 (boundary of the disc) are its two famous corollaries.

3.41 Theorem. *Given two sequences (a_n) and (b_n) , put*

$$A_n = \sum_{k=0}^n a_k$$

if $n \geq 0$; put $A_{-1} = 0$. Then, if $0 \leq p \leq q$, we have

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \quad (3.20)$$

Proof.

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^q (A_n - A_{n-1}) b_n = \sum_{n=p}^q A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1},$$

and the last expression on the right is clearly equal to the right side of (3.20). $\square$

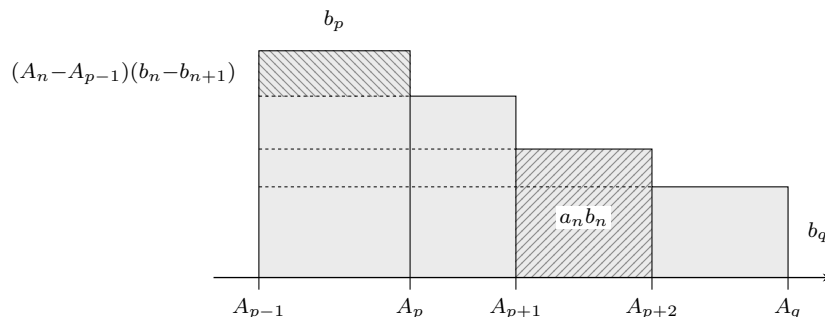
Formula (3.20), the so-called “partial summation formula,” is useful in the investigation of series of the form $\sum a_n b_n$, particularly when (b_n) is monotonic. We shall now give applications.

Integration by parts, one chapter early

Rename the pieces and (3.20) is an old friend: A_n is the “antiderivative” of a_n (partial sums undo differencing, since $a_n = A_n - A_{n-1}$), and $b_n - b_{n+1}$ is the “derivative” of b_n . The formula then reads $\sum a b = - \sum A \Delta b$ + boundary terms, which is $\int u dv = uv - \int v du$ with sums for integrals. Chapter 6 proves the integral version as 6.22, by essentially this computation applied to Riemann–Stieltjes sums.

The strategic content is the same in both settings: move the burden from the factor you cannot control onto the factor that varies slowly.

Here Σa_n may diverge, but only the *bounded* A_n appear on the right, multiplied by the increments of a monotone b_n — and those increments telescope.



Both sides of (3.20) measure this one staircase: columns of height b_n stand over the increments $a_n = A_n - A_{n-1}$. Sliced vertically (hatched sample), the area is $\sum_p^q a_n b_n$. Sliced horizontally, the slab between levels b_{n+1} and b_n has width $A_n - A_{p-1}$, and the base slab, height b_q , runs the full width $A_q - A_{p-1}$; expanding the widths telescopes all the A_{p-1} 's into the single boundary term $-A_{p-1}b_p$ of (3.20). Drawn for $a_n > 0$ and decreasing positive b_n — exactly 3.42's hypotheses; Rudin's two-line proof is the same bookkeeping with signs allowed.

3.42 Theorem. Suppose

- (a) the partial sums A_n of Σa_n form a bounded sequence;
- (b) $b_0 \geq b_1 \geq b_2 \geq \cdots$;
- (c) $\lim_{n \rightarrow \infty} b_n = 0$.

Then $\Sigma a_n b_n$ converges.

Proof. Choose M such that $|A_n| \leq M$ for all n . Given $\varepsilon > 0$, there is an integer N such that $b_N \leq \varepsilon/(2M)$. For $N \leq p \leq q$, we have

$$\begin{aligned} \left| \sum_{n=p}^q a_n b_n \right| &= \left| \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p \right| \\ &\leq M \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + b_q + b_p \right] \\ &= 2M b_p \leq 2M b_N \leq \varepsilon. \end{aligned}$$

37 The telescope is the payoff: the increments $b_n - b_{n+1}$ are nonnegative, so absolute values cost nothing, and their sum collapses to $b_p - b_q$. All the oscillation of Σa_n has been absorbed into the single constant M ; what remains converges because (b_n) does. Note the shape of the conclusion: convergence via 3.22, no sum ever named.

Convergence now follows from the Cauchy criterion. We note that the first inequality in the above chain depends of course on the fact that $b_n - b_{n+1} \geq 0$.³⁷ $\square$

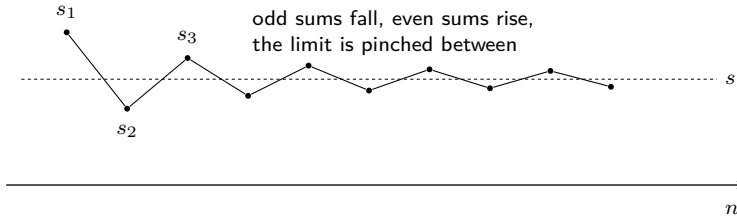
3.43 Theorem. Suppose

- (a) $|c_1| \geq |c_2| \geq |c_3| \geq \cdots$;
- (b) $c_{2m-1} \geq 0, c_{2m} \leq 0$ ($m = 1, 2, 3, \dots$);
- (c) $\lim_{n \rightarrow \infty} c_n = 0$.

Then $\sum c_n$ converges.

Series for which (b) holds are called “alternating series”; the theorem was known to Leibnitz.

Proof. Apply Theorem 3.42 with $a_n = (-1)^{n+1}$, $b_n = |c_n|$.³⁸ □



Alternating series: each swing overshoots the limit and is shorter than the last. The even partial sums increase, the odd ones decrease, and $|s - s_n| \leq |c_{n+1}|$ — the error is at most the first omitted term.

3.44 Theorem. Suppose the radius of convergence of $\sum c_n z^n$ is 1, and suppose $c_0 \geq c_1 \geq c_2 \geq \dots$, $\lim_{n \rightarrow \infty} c_n = 0$. Then $\sum c_n z^n$ converges at every point on the circle $|z| = 1$, except possibly at $z = 1$.

Proof. Put $a_n = z^n$, $b_n = c_n$. The hypotheses of Theorem 3.42 are then satisfied, since

$$|A_n| = \left| \sum_{m=0}^n z^m \right| = \left| \frac{1 - z^{n+1}}{1 - z} \right| \leq \frac{2}{|1 - z|},$$

if $|z| = 1$ and $z \neq 1$.³⁹ □

ABSOLUTE CONVERGENCE

The series $\sum a_n$ is said to *converge absolutely* if the series $\sum |a_n|$ converges.

3.45 Theorem. If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. The assertion follows from the inequality

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k|,$$

plus the Cauchy criterion. □

3.46 Remarks. For series of positive terms, absolute convergence is the same as convergence.

If $\sum a_n$ converges, but $\sum |a_n|$ diverges, we say that $\sum a_n$ converges *conditionally* [[nonabsolutely]].⁴⁰ For instance, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges conditionally (Theorem 3.43).

38 The partial sums of $(-1)^{n+1}$ are $1, 0, 1, 0, \dots$ — bounded by $M = 1$ — and the $|c_n|$ decrease to 0. The most famous instance: $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges — its sum turns out to be $\log 2$, a fact that needs Chapter 8 — while its absolute version, the harmonic series, diverges. The gap between those two facts is the subject of the whole next section.

39 The bound degrades as z approaches $1 - |1 - z|$ in the denominator — which is why $z = 1$ must be excepted; there $A_n = n + 1$ is unbounded and 3.42 has nothing to grip. This settles 3.40(d): $\sum z^n/n$ converges everywhere on the unit circle except at $z = 1$, where it is the harmonic series. Alternating series are the case $z = -1$.

40 The doubled brackets are the retyped edition's editorial mark, explained in its foreword: Rudin's own word was “nonabsolutely,” silently modernised to “conditionally” throughout, with the original recorded once — here — at the point of first use.

The comparison test, as well as the root and ratio tests, is really a test for absolute convergence, and therefore cannot give any information about conditionally convergent series. Summation by parts can sometimes be used to handle the latter. In particular, power series converge absolutely in the interior of the circle of convergence.

We shall see that we may operate with absolutely convergent series very much as with finite sums. We may multiply them term by term and we may change the order in which the additions are carried out, without affecting the sum of the series. But for conditionally convergent series this is no longer true, and more care has to be taken when dealing with them.⁴¹

41 This paragraph is the thesis statement for the rest of the chapter. “Absolutely convergent” will turn out to mean “behaves like a finite sum” — products work (3.50), rearrangements are harmless (3.55) — while “conditionally convergent” means the sum is an artifact of the order of summation, so much so that rearrangement can produce any sum whatever (3.54). Absolute convergence is not a technicality; it is the boundary between arithmetic and analysis.

ADDITION AND MULTIPLICATION OF SERIES

WHICH FINITE-SUM RULES SURVIVE THE LIMIT | The chapter’s closing question: which of the ordinary rules of arithmetic survive when “sum” means “limit of partial sums”? Addition survives unconditionally (3.47). Multiplication survives if at least one factor converges absolutely (Mertens, 3.50), can fail for two conditionally convergent factors (3.49), yet can never converge to the *wrong* value (Abel, 3.51). Rearrangement — the commutative law — is the most fragile of all, and the next section is devoted to its spectacular failure.

3.47 Theorem. If $\sum a_n = A$ and $\sum b_n = B$, then $\sum(a_n + b_n) = A + B$, and $\sum ca_n = cA$, for any fixed c .

Proof. Let

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k.$$

Then

$$A_n + B_n = \sum_{k=0}^n (a_k + b_k).$$

Since $\lim_{n \rightarrow \infty} A_n = A$ and $\lim_{n \rightarrow \infty} B_n = B$, we see that

$$\lim_{n \rightarrow \infty} (A_n + B_n) = A + B.$$

The proof of the second assertion is even simpler. □

Thus two convergent series may be added term by term, and the resulting series converges to the sum of the two series. The situation becomes more complicated when we consider multiplication of two series. To begin with, we have to define the product. This can be done in several ways; we shall consider the so-called “Cauchy product.”

3.48 Definition. Given Σa_n and Σb_n , we put

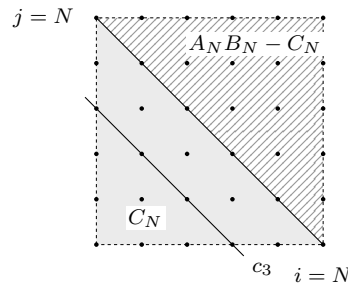
$$c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots)$$

and call Σc_n the *product* of the two given series.

This definition may be motivated as follows. If we take two power series $\Sigma a_n z^n$ and $\Sigma b_n z^n$, multiply them term by term, and collect terms containing the same power of z , we get

$$\begin{aligned} \sum_{n=0}^{\infty} a_n z^n \cdot \sum_{n=0}^{\infty} b_n z^n &= (a_0 + a_1 z + a_2 z^2 + \dots)(b_0 + b_1 z + b_2 z^2 + \dots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)z \\ &\quad + (a_0 b_2 + a_1 b_1 + a_2 b_0)z^2 + \dots \\ &= c_0 + c_1 z + c_2 z^2 + \dots \end{aligned}$$

Setting $z = 1$, we arrive at the above definition.⁴²



42 The c_n collect the products $a_k b_j$ along the anti-diagonals $k + j = n$ of the multiplication array — the same anti-diagonals as in the countability proof 2.12, and for the same reason: each one is finite. The definition is rigged for power series, and 3.51's proof will return there.

The multiplication array $a_i b_j$, one dot per product. Each c_n strings an anti-diagonal (shown: $i + j = 3$), and every anti-diagonal is finite — the definition's whole content. The shaded triangle $i + j \leq N$ sums to C_N ; the dashed square sums to $A_N B_N$. Everything about Σc_n versus AB is the fate of the hatched difference: Mertens (3.50) shows absolute convergence of one factor kills it, while in Example 3.49 it carries mass ≈ 2 forever. These are the anti-diagonals of 2.12's walk, finite for the same reason.

3.49 Example. If

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k,$$

and $A_n \rightarrow A$, $B_n \rightarrow B$, then it is not at all clear that (C_n) will converge to AB , since we do not have $C_n = A_n B_n$. The dependence of (C_n) on (A_n) and (B_n) is quite a complicated one (see the proof of Theorem 3.50). We shall now show that the product of two convergent series may actually diverge.

The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$$

converges (Theorem 3.43). We form the product of this series with itself and obtain

$$\begin{aligned} \sum_{n=0}^{\infty} c_n &= 1 - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}\sqrt{2}} + \frac{1}{\sqrt{3}} \right) \\ &\quad - \left(\frac{1}{\sqrt{4}} + \frac{1}{\sqrt{3}\sqrt{2}} + \frac{1}{\sqrt{2}\sqrt{3}} + \frac{1}{\sqrt{4}} \right) + \cdots, \end{aligned}$$

so that

$$c_n = (-1)^n \sum_{k=0}^n \frac{1}{\sqrt{(n-k+1)(k+1)}}.$$

Since

$$(n-k+1)(k+1) = \left(\frac{n}{2} + 1 \right)^2 - \left(\frac{n}{2} - k \right)^2 \leq \left(\frac{n}{2} + 1 \right)^2,$$

we have

$$|c_n| \geq \sum_{k=0}^n \frac{2}{n+2} = \frac{2(n+1)}{n+2},$$

so that the condition $c_n \rightarrow 0$, which is necessary for the convergence of $\sum c_n$, is not satisfied.⁴³

In view of the next theorem, due to Mertens, we note that we have here considered the product of two conditionally convergent series.

3.50 Theorem. *Suppose*

- (a) $\sum_{n=0}^{\infty} a_n$ converges absolutely,
- (b) $\sum_{n=0}^{\infty} a_n = A$,
- (c) $\sum_{n=0}^{\infty} b_n = B$,
- (d) $c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots).$

Then

$$\sum_{n=0}^{\infty} c_n = AB.$$

That is, the product of two convergent series converges, and to the right value, if at least one of the two series converges absolutely.

Proof. Put

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k, \quad \beta_n = B_n - B.$$

Then

$$\begin{aligned} C_n &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \cdots + a_n (B + \beta_0) \\ &= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \cdots + a_n \beta_0. \end{aligned}$$

43 Why it fails: each c_n averages $n+1$ products of terms of size about $1/\sqrt{n}$ each, so the products are of size about $1/n$ and their sum stays near 2. The factor-series converge only by cancellation, and the Cauchy product lines up terms so that the cancellation is destroyed. Convergence-by-cancellation is not robust under rearrangement-like operations — the theme of the whole final section.

44 The first rewriting is the anti-diagonal picture again: summing $c_0, \dots, c_n$ sweeps the triangle $k+j \leq n$ of the array, which can be sliced as a_k times the partial sum B_{n-k} . Then each B_{n-k} is split into limit plus error, $B + \beta_{n-k}$, isolating the main term $A_n B$. The proof will live or die on the error convolution γ_n .

44 Put

$$\gamma_n = a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0.$$

We wish to show that $C_n \rightarrow AB$. Since $A_nB \rightarrow AB$, it suffices to show that

$$\lim_{n \rightarrow \infty} \gamma_n = 0. \quad (3.21)$$

Put

$$\alpha = \sum_{n=0}^{\infty} |a_n|.$$

[It is here that we use (a).] Let $\varepsilon > 0$ be given. By (c), $\beta_n \rightarrow 0$. Hence we can choose N such that $|\beta_n| \leq \varepsilon$ for $n \geq N$, in which case

$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N-1} + \cdots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \alpha. \end{aligned}$$

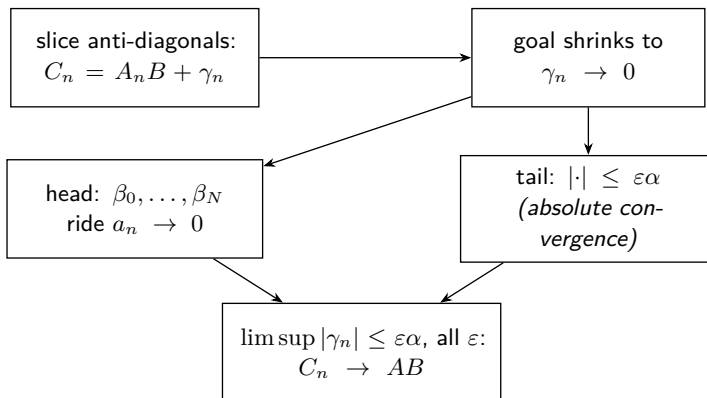
Keeping N fixed and letting $n \rightarrow \infty$, we get

$$\limsup_{n \rightarrow \infty} |\gamma_n| \leq \varepsilon \alpha,$$

since $a_k \rightarrow 0$ as $k \rightarrow \infty$. Since ε is arbitrary, (3.21) follows. ⁴⁵

□

45 The split-the-sum device, and the same two-parameter discipline as in 3.31: the old errors $\beta_0, \dots, \beta_N$ are finitely many and ride multiples of $a_n \rightarrow 0$; the recent errors are small and ride the absolutely convergent tail, total at most $\varepsilon\alpha$. Freeze N , let n run, then shrink ε . Without (a) the constant α does not exist, and 3.49 shows the theorem genuinely fails.



Mertens as a flow diagram: isolate the error convolution, split it at N , and let each half die for its own reason. Hypothesis (a) funds exactly one box — the tail.

Another question which may be asked is whether the series Σc_n , if convergent, must have the sum AB . Abel showed that the answer is in the affirmative.

3.51 Theorem. *If the series Σa_n , Σb_n , Σc_n converge to A , B , C , and $c_n = a_0 b_n + \cdots + a_n b_0$, then $C = AB$.*

Here no assumption is made concerning absolute convergence. We shall give a simple proof (which depends on the continuity of power series) after Theorem 8.2. ⁴⁶

REARRANGEMENTS

46 A promissory note worth remembering when it is paid. The future proof: the three power series $\Sigma a_n x^n$, $\Sigma b_n x^n$, $\Sigma c_n x^n$ converge absolutely for $0 < x < 1$, where Mertens applies and gives $f(x)g(x) = h(x)$; then let $x \rightarrow 1$ and invoke Abel's continuity theorem (8.2) three times. The pattern — prove an identity where convergence is absolute, then reach the boundary by continuity — is a first taste of a standard analytic method.

3.52 Definition. Let (k_n) , $n = 1, 2, 3, \dots$, be a sequence in which every positive integer appears once and only once (that is, (k_n) is a 1-1 function from $\mathbb{N}$ onto $\mathbb{N}$, in the notation of Definition 2.2). Putting

$$a'_n = a_{k_n} \quad (n = 1, 2, 3, \dots),$$

we say that $\Sigma a'_n$ is a *rearrangement* of Σa_n .

If (s_n) , (s'_n) are the sequences of partial sums of Σa_n , $\Sigma a'_n$, it is easily seen that, in general, these two sequences consist of entirely different numbers. We are thus led to the problem of determining under what conditions all rearrangements of a convergent series will converge, and whether the sums are necessarily the same.

3.53 Example. Consider the convergent series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots \quad (3.22)$$

and one of its rearrangements

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots \quad (3.23)$$

in which two positive terms are always followed by one negative. If s is the sum of (3.22), then

$$s < 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

⁴⁷ Since

$$\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} > 0$$

for $k \geq 1$, we see that $s'_3 < s'_6 < s'_9 < \cdots$, where s'_n is the n th partial sum of (3.23).

Hence

$$\limsup_{n \rightarrow \infty} s'_n > s'_3 = \frac{5}{6},$$

so that (3.23) certainly does not converge to s [we leave it to the reader to verify that (3.23) does, however, converge].⁴⁸

This example illustrates the following theorem, due to Riemann.

3.54 Theorem. Let Σa_n be a series of real numbers which converges, but not absolutely. Suppose

$$-\infty \leq \alpha \leq \beta \leq +\infty.$$

Then there exists a rearrangement $\Sigma a'_n$ with partial sums s'_n such that

$$\liminf_{n \rightarrow \infty} s'_n = \alpha, \quad \limsup_{n \rightarrow \infty} s'_n = \beta. \quad (3.24)$$

Proof. Let

$$p_n = \frac{|a_n| + a_n}{2}, \quad q_n = \frac{|a_n| - a_n}{2} \quad (n = 1, 2, 3, \dots).$$

⁴⁷ Why this bound: (3.22) is alternating with decreasing terms, so by the zigzag picture at 3.43 the partial sums oscillate around s , and every odd partial sum overestimates it — $s < s_3 = 5/6$.

⁴⁸ The same numbers, added in a different order, give a different sum: taking positives twice as fast as negatives borrows against the divergent positive half, and each three-term block adds a surplus. This is the concrete disaster; 3.54 shows it is total.

Then $p_n - q_n = a_n$, $p_n + q_n = |a_n|$, $p_n \geq 0$, $q_n \geq 0$. The series Σp_n , Σq_n must both diverge, for if both were convergent, then

$$\Sigma(p_n + q_n) = \Sigma|a_n|$$

would converge, contrary to hypothesis. Since

$$\sum_{n=1}^N a_n = \sum_{n=1}^N (p_n - q_n) = \sum_{n=1}^N p_n - \sum_{n=1}^N q_n,$$

divergence of Σp_n and convergence of Σq_n (or vice versa) implies divergence of Σa_n , again contrary to hypothesis.⁴⁹

Now let $P_1, P_2, P_3, \dots$ denote the nonnegative terms of Σa_n , in the order in which they occur, and let $Q_1, Q_2, Q_3, \dots$ be the absolute values of the negative terms of Σa_n , also in their original order.

The series ΣP_n , ΣQ_n differ from Σp_n , Σq_n only by zero terms, and are therefore divergent.

We shall construct sequences (m_n) , (k_n) such that the series

$$\begin{aligned} P_1 + \dots + P_{m_1} - Q_1 - \dots - Q_{k_1} + P_{m_1+1} + \dots \\ + P_{m_2} - Q_{k_1+1} - \dots - Q_{k_2} + \dots, \end{aligned} \quad (3.25)$$

which clearly is a rearrangement of Σa_n , satisfies (3.24).

Choose real-valued sequences (α_n) , (β_n) such that $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$, $\alpha_n < \beta_n$, $\beta_1 > 0$.

Let m_1, k_1 be the smallest positive integers such that

$$\begin{aligned} P_1 + \dots + P_{m_1} &> \beta_1, \\ P_1 + \dots + P_{m_1} - Q_1 - \dots - Q_{k_1} &< \alpha_1; \end{aligned}$$

let m_2, k_2 be the smallest positive integers such that

$$\begin{aligned} P_1 + \dots + P_{m_1} - Q_1 - \dots - Q_{k_1} + P_{m_1+1} + \dots + P_{m_2} &> \beta_2, \\ P_1 + \dots + P_{m_1} - Q_1 - \dots - Q_{k_1} + P_{m_1+1} + \dots + P_{m_2} \\ - Q_{k_1+1} - \dots - Q_{k_2} &< \alpha_2, \end{aligned}$$

and continue in this way. This is possible since ΣP_n and ΣQ_n diverge.

If x_n, y_n denote the partial sums of (3.25) whose last terms are P_{m_n} , $-Q_{k_n}$, then

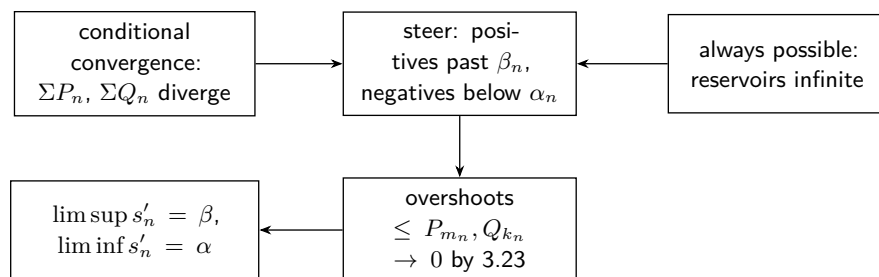
$$|x_n - \beta_n| \leq P_{m_n}, \quad |y_n - \alpha_n| \leq Q_{k_n}.$$

Since $P_n \rightarrow 0$ and $Q_n \rightarrow 0$ as $n \rightarrow \infty$, we see that $x_n \rightarrow \beta$, $y_n \rightarrow \alpha$.⁵⁰

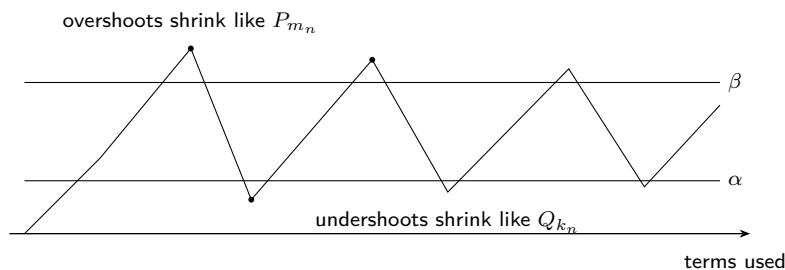
Finally, it is clear that no number less than α or greater than β can be a subsequential limit of the partial sums (3.25). $\square$

49 The anatomy of conditional convergence, and the engine of the whole proof: split the series into its positive part and its negative part, and both halves diverge. The finite sum s is the difference of two infinities kept in balance by the original ordering. That balance is the only thing a rearrangement can destroy — and it can destroy it completely, because each half is an inexhaustible reservoir.

50 The overshoot estimate is where convergence of the original series earns its keep: $a_n \rightarrow 0$ (3.23), so the reservoir terms P_n, Q_n shrink, and each crossing of the target overshoots by at most the last term used. The steering wheel: pay out positives until just past β_n , then negatives until just below α_n — “smallest integer such that” makes each overshoot minimal.



Riemann's theorem as a flow diagram: divergence of both reservoirs makes the steering possible, convergence of the original series ($a_n \rightarrow 0$) makes it accurate. Each hypothesis funds exactly one arrow.



Riemann's steering: positives push the partial sums just past β , negatives pull them just below α , forever. Divergence of both reservoirs keeps the steering possible; $a_n \rightarrow 0$ makes the overshoots die, so the upper and lower limits are exactly β and α . Take $\alpha = \beta$ to hit any prescribed sum.

3.55 Theorem. *If Σa_n is a series of complex numbers which converges absolutely, then every rearrangement of Σa_n converges, and they all converge to the same sum.*

Proof. Let $\Sigma a'_n$ be a rearrangement, with partial sums s'_n . Given $\varepsilon > 0$, there exists an integer N such that $m \geq n \geq N$ implies

$$\sum_{i=n}^m |a_i| \leq \varepsilon. \quad (3.26)$$

Now choose p such that the integers $1, 2, \dots, N$ are all contained in the set $k_1, k_2, \dots, k_p$ (we use the notation of Definition 3.52). Then if $n > p$, the numbers $a_1, \dots, a_N$ will cancel in the difference $s_n - s'_n$, so that $|s_n - s'_n| \leq \varepsilon$, by (3.26). Hence (s'_n) converges to the same sum as (s_n) .⁵¹ □

51 Absolute convergence makes the tail unconditionally negligible: (3.26) bounds any finite batch of far-out terms regardless of order or sign, so once both partial sums have swallowed $a_1, \dots, a_N$, they differ only by scraps from the tail. Compare 3.54, where the tail's two halves were infinite reservoirs; here the tail is just small. This is the precise sense of Rudin's slogan at 3.46 that absolutely convergent series behave like finite sums.

CHAPTER 3 IN ONE PARAGRAPH

Convergence of a sequence is defined by the ε - N contract (3.1), and completeness makes it checkable from inside: Cauchy in $\mathbb{R}^k$ means convergent (3.11), monotone-and-bounded means convergent (3.14), and every real sequence has lim sup and lim inf as its extreme subsequential limits (3.17). A series is the sequence of

its partial sums (3.21), so those three facts become the Cauchy criterion (3.22), the boundedness criterion for nonnegative series (3.24), and the root test (3.33); every test is ultimately a comparison (3.25) with the geometric series (3.26), except summation by parts (3.41–3.44), which alone reaches conditional convergence. The number e arrives by series, fast enough for irrationality to fall out (3.30–3.32). The chapter’s final lesson is the boundary drawn at 3.46 and dramatised at 3.54–3.55: absolutely convergent series behave like finite sums under products and rearrangements; conditionally convergent series have sums that depend on the order of the terms — so much so that any prescribed value, or none, can be arranged.

EXERCISES

HOW TO USE THESE

Chapter 3’s exercise set contains two of the most important constructions in the book. 3.23–3.24 build the **completion** of an arbitrary metric space out of Cauchy sequences — Tao’s construction of $\mathbb{R}$ itself, carried out here as an exercise, with 3.25 closing the circle by asking what the completion of $\mathbb{Q}$ is. 3.22 is **Baire’s theorem** in its general form, promised at Exercise 2.30.

Technique builders. 3.3 and 3.16–3.18 are the monotone-convergence workhorses: recursive sequences squeezed by 3.14, with 3.16 being Newton’s method for square roots and its blazing quadratic convergence. 3.14 develops Cesàro means — the first taste of summability theory, needed for Fourier series (8.15–8.16) and Tauberian arguments.

Counterexample duty. 3.6–3.8 and 3.11–3.12 sharpen the convergence tests; 3.11(b) and 3.12(b) are the pairs Rudin cited in the text to kill the “boundary between convergence and divergence” conjecture.

Routine. 3.1, 3.2, 3.9, 3.20 are quick checks; do them first.

3.1. Prove that convergence of (s_n) implies convergence of $(|s_n|)$. Is the converse true?⁵²

3.2. Calculate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$.⁵³

3.3. If $s_1 = \sqrt{2}$ and

$$s_{n+1} = \sqrt{2 + \sqrt{s_n}} \quad (n = 1, 2, 3, \dots),$$

52 Reverse triangle inequality (Exercise 1.13): $||s_n| - |s|| \leq |s_n - s|$. The converse fails at once: $s_n = (-1)^n$.

53 Multiply and divide by the conjugate $\sqrt{n^2 + n} + n$; the answer is $\frac{1}{2}$. The moral: $\infty - \infty$ expressions are settled by algebra first, limit theorems second — 3.3 applies only after the difference is rewritten as a quotient.

54 The 3.14 template, used again in 3.16–3.18: prove boundedness ($s_n < 2$) and monotonicity together by induction, then monotone convergence hands you a limit. Only afterwards may you pass to the limit in the recursion to identify it — existence must precede the algebra.

55 Track the two subsequences separately: the even-indexed terms crawl up toward $\frac{1}{2}$, the odd ones toward 1. So $E = \{\frac{1}{2}, 1\}$, $\liminf = \frac{1}{2}$, $\limsup = 1$ — a sequence need not oscillate symmetrically about anything.

56 Inequality, not equality — and strictness is common: $a_n = (-1)^n$, $b_n = (-1)^{n+1}$ gives $0 < 2$. The two sequences need not peak at the same indices. (The excluded case is exactly the arithmetic left undefined at 1.23.) The working proof uses the tail-sup form from the box at 3.17:

$\sup_{n \geq N} (a_n + b_n) \leq \sup_{n \geq N} a_n + \sup_{n \geq N} b_n$, then let $N \rightarrow \infty$.

57 (a) telescopes to $\sqrt{N+1} - 1$: diverges. (b) conjugate trick gives $a_n \sim \frac{1}{2}n^{-3/2}$: converges by comparison with 3.28. (c) root test: $\sqrt[n]{a_n} = \sqrt[n]{n} - 1 \rightarrow 0$ by 3.20(c): converges. (d) splits by $|z|$: for $|z| > 1$ comparison with $\Sigma|z|^{-n}$ converges; for $|z| \leq 1$ the terms do not tend to 0 (3.23) — watch $|z| = 1$, where $1 + z^n$ can even vanish.

58 The trick to remember:

$\sqrt{a_n}/n \leq \frac{1}{2}(a_n + 1/n^2)$ by the AM–GM inequality $xy \leq \frac{1}{2}(x^2 + y^2)$ — which is Schwarz 1.35 for $n = 1$. Both halves converge, so comparison finishes it.

59 Summation by parts, but not quite 3.42, whose hypothesis $b_n \rightarrow 0$ may fail here. Write $b_n = (b_n - b) + b$ where $b = \lim b_n$ (exists by 3.14): the b -part converges by 3.47, and $(b_n - b)$ decreases (or increases) to 0, so 3.42 handles the rest. Note the partial sums A_n are bounded because Σa_n converges — a weaker demand than 3.42 makes.

60 All four by 3.37's byproduct ($\lim c_{n+1}/c_n$, when it exists, is α): (a) $R = 1$ — polynomial factors never move R , by 3.20(c); (b) $R = \infty$; (c) $R = \frac{1}{2}$; (d) $R = 3$.

prove that (s_n) converges, and that $s_n < 2$ for $n = 1, 2, 3, \dots$.⁵⁴

3.4. Find the upper and lower limits of the sequence (s_n) defined by

$$s_1 = 0; \quad s_{2m} = \frac{s_{2m-1}}{2}; \quad s_{2m+1} = \frac{1}{2} + s_{2m}.$$

55

3.5. For any two real sequences (a_n) , (b_n) , prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n,$$

provided the sum on the right is not of the form $\infty - \infty$.⁵⁶

3.6. Investigate the behavior (convergence or divergence) of Σa_n if

(a) $a_n = \sqrt{n+1} - \sqrt{n}$;

(b) $a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$;

(c) $a_n = (\sqrt[n]{n} - 1)^n$;

(d) $a_n = \frac{1}{1 + z^n}$, for complex values of z .

57

3.7. Prove that the convergence of Σa_n implies the convergence of

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n},$$

if $a_n \geq 0$.⁵⁸

3.8. If Σa_n converges and if (b_n) is monotonic and bounded, prove that $\Sigma a_n b_n$ converges.⁵⁹

3.9. Find the radius of convergence of each of the following power series:

(a) $\sum n^3 z^n$,

(c) $\sum \frac{2^n}{n^2} z^n$,

(b) $\sum \frac{2^n}{n!} z^n$,

(d) $\sum \frac{n^3}{3^n} z^n$.

60

3.10. Suppose that the coefficients of the power series $\Sigma a_n z^n$ are integers, infinitely many of which are distinct from zero. Prove that the radius of convergence is at most 1.⁶¹

3.11. Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and Σa_n diverges.

(a) Prove that $\sum \frac{a_n}{1 + a_n}$ diverges.

(b) Prove that

$$\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} \geq 1 - \frac{s_N}{s_{N+k}}$$

and deduce that $\sum \frac{a_n}{s_n}$ diverges.

(c) Prove that

$$\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$$

and deduce that $\sum \frac{a_n}{s_n^2}$ converges.

(d) What can be said about

$$\sum \frac{a_n}{1 + na_n} \quad \text{and} \quad \sum \frac{a_n}{1 + n^2 a_n} ?$$

62

3.12. Suppose $a_n > 0$ and Σa_n converges. Put

$$r_n = \sum_{m=n}^{\infty} a_m.$$

(a) Prove that

$$\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$$

if $m < n$, and deduce that $\sum \frac{a_n}{r_n}$ diverges.

(b) Prove that

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

and deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

63

3.13. Prove that the Cauchy product of two absolutely convergent series converges absolutely.⁶⁴

3.14. If (s_n) is a complex sequence, define its arithmetic means σ_n by

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1} \quad (n = 0, 1, 2, \dots).$$

(a) If $\lim s_n = s$, prove that $\lim \sigma_n = s$.

61 A nonzero integer has $|a_n| \geq 1$, so $\sqrt[n]{|a_n|} \geq 1$ infinitely often and $\alpha = \limsup \geq 1$. Integrality forces smallness to be impossible — the same arithmetic leverage as in the irrationality proof 3.32.

62 This is the divergent half of Rudin's "no boundary" claim after 3.29: by (b), $\Sigma a_n/s_n$ diverges yet its terms are smaller than a_n by the unbounded factor s_n — every divergent series dominates a slower one. Part (b)'s display is a Cauchy-criterion violation served ready-made: fix N , let k grow, and the blocks refuse to shrink because $s_{N+k} \rightarrow \infty$. Part (c) telescopes. In (d) the first can do either (try $a_n = 1$ vs. a_n supported on sparse blocks); the second always converges, since $a_n/(1 + n^2 a_n) \leq 1/n^2$.

63 The convergent half of "no boundary": by (b), dividing a convergent series by the vanishing factor $\sqrt{r_n}$ leaves it convergent — every convergent series is dominated by a slower one. Together with 3.11(b) this kills the conjecture in both directions. The engine in both parts is the same: r_n telescopes ($a_n = r_n - r_{n+1}$), and (b) is the algebraic identity $\frac{r_n - r_{n+1}}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$, i.e. $\sqrt{r_n} + \sqrt{r_{n+1}} < 2\sqrt{r_n}$.

64 Compare $\sum_n |c_n|$ against $(\Sigma|a_n|)(\Sigma|b_n|)$: every partial sum of $|c_n|$ is trapped in a triangle of the array of $|a_k b_j|$, which sits inside a full rectangle. With 3.50 this closes the multiplication story: absolute $\times$ absolute is absolute; absolute $\times$ conditional converges to the right value (Mertens); conditional $\times$ conditional can diverge (3.49).

- (b) Construct a sequence (s_n) which does not converge, although $\lim \sigma_n = 0$.
- (c) Can it happen that $s_n > 0$ for all n and that $\limsup s_n = \infty$, although $\lim \sigma_n = 0$?
- (d) Put $a_n = s_n - s_{n-1}$ for $n \geq 1$. Show that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^n k a_k.$$

Assume that $\lim(na_n) = 0$ and that (σ_n) converges. Prove that (s_n) converges. [This gives a converse of (a), but under the additional assumption that $na_n \rightarrow 0$.]

- (e) Derive the last conclusion from a weaker hypothesis: Assume $M < \infty$, $|na_n| \leq M$ for all n , and $\lim \sigma_n = \sigma$. Prove that $\lim s_n = \sigma$, by completing the following outline:

If $m < n$, then

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^n (s_n - s_i).$$

For these i ,

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$ and associate with each n the integer m that satisfies

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then $(m+1)/(n-m) \leq 1/\varepsilon$ and $|s_n - s_i| < M\varepsilon$. Hence

$$\limsup_{n \rightarrow \infty} |s_n - \sigma| \leq M\varepsilon.$$

Since ε was arbitrary, $\lim s_n = \sigma$.

65

65 Cesàro summability, and a first look at an entire subject. Part (a) says averaging never destroys a limit; (b), with $s_n = (-1)^n$, says it can manufacture one — so “Cesàro sum” genuinely extends “sum,” which is how Fourier series will be rescued at 8.15–8.16 (Fejér’s idea). Parts (d) and (e) are Tauberian theorems: conditions under which the extension collapses back to ordinary convergence. Hardy’s condition $|na_n| \leq M$ in (e) is the historical original. The two-parameter bookkeeping in the outline — m tied to n through ε — is 3.31’s freeze-and-release discipline pushed one level up.

66 The checklist of which theory is metric and which is order-theoretic: everything driven by the Cauchy criterion and the triangle inequality survives in $\mathbb{R}^k$ verbatim. What is missing from the list matters too — 3.24, 3.27, and comparison (b) need nonnegativity, and the root/ratio tests survive only because they test the scalar series $\sum |\mathbf{a}_n|$.

3.15. Definition 3.21 can be extended to the case in which the a_n lie in some fixed $\mathbb{R}^k$. Absolute convergence is defined as convergence of $\sum |\mathbf{a}_n|$. Show that Theorems 3.22, 3.23, 3.25(a), 3.33, 3.34, 3.42, 3.45, 3.47, and 3.55 are true in this more general setting. (Only slight modifications are required in any of the proofs.)⁶⁶

3.16. Fix a positive number α . Choose $x_1 > \sqrt{\alpha}$, and define $x_2, x_3, x_4, \dots$ by the recursion formula

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right).$$

(a) Prove that (x_n) decreases monotonically and that $\lim x_n = \sqrt{\alpha}$.

(b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$, and show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}},$$

so that, setting $\beta = 2\sqrt{\alpha}$,

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n} \quad (n = 1, 2, 3, \dots).$$

(c) This is a good algorithm for computing square roots, since the recursion formula is simple and the convergence is extremely rapid. For example, if $\alpha = 3$ and $x_1 = 2$, show that $\varepsilon_1/\beta < 1/10$ and that therefore

$$\varepsilon_5 < 4 \cdot 10^{-16}, \quad \varepsilon_6 < 4 \cdot 10^{-32}.$$

67

3.17. Fix $\alpha > 1$. Take $x_1 > \sqrt{\alpha}$, and define

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}.$$

(a) Prove that $x_1 > x_3 > x_5 > \dots$.

(b) Prove that $x_2 < x_4 < x_6 < \dots$.

(c) Prove that $\lim x_n = \sqrt{\alpha}$.

(d) Compare the rapidity of convergence of this process with the one described in Exercise 3.16.

68

3.18. Replace the recursion formula of Exercise 3.16 by

$$x_{n+1} = \frac{p-1}{p}x_n + \frac{\alpha}{p}x_n^{-p+1},$$

where p is a fixed positive integer, and describe the behavior of the resulting sequences (x_n) .⁶⁹

3.19. Associate to each sequence $a = (\alpha_n)$, in which α_n is 0 or 2, the real number

$$x(a) = \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n}.$$

Prove that the set of all $x(a)$ is precisely the Cantor set described in Theorem 2.44.⁷⁰

67 Newton's method on $f(x) = x^2 - \alpha$, and the recursion Chapter 1's note on Example 1.1 promised: it always lands above $\sqrt{\alpha}$ (AM-GM), so the sequence is decreasing and 3.14 applies. Part (b) is quadratic convergence — the error squares each step, doubling the correct digits — against which the linear rate of 1.1's map $q = (2p+2)/(p+2)$, and of 3.17 below, can be measured.

68 The general- α version of Chapter 1's map (there $\alpha = 2$: $q = (2+p)/(1+p)$). Unlike Newton, it alternates sides of $\sqrt{\alpha}$, so the two subsequences must be handled separately by 3.14 and then reunited — the strategy of 3.4. Convergence is linear: the error is multiplied by roughly $(\sqrt{\alpha}-1)/(\sqrt{\alpha}+1)$ each step, one new digit per few iterations, versus Newton's doubling.

69 Newton for $f(x) = x^p - \alpha$: the limit is $\sqrt[p]{\alpha}$, with the same quadratic convergence and the same monotone-decrease-from-above proof. A numerical companion to Theorem 1.21, which proved the p th root exists; this computes it.

70 The address system of Chapter 2's binary-tree figure made exact: digit α_n chooses the left or right third at stage n , and the series is the point the nested thirds close on. Series notation turns the geometric description into a formula — and makes the bijection with $\{0, 2\}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}}$ (hence uncountability, 2.14) immediate.

71 Split $d(p_n, p) \leq d(p_n, p_{n_i}) + d(p_{n_i}, p)$: the first half is small because the sequence is Cauchy, the second because the subsequence converges. Small but used constantly — including in the very next two exercises, and it is how 3.11(b) could have been proved from 3.6(a) directly.

- 3.20.** Suppose (p_n) is a Cauchy sequence in a metric space X , and some subsequence (p_{n_i}) converges to a point $p \in X$. Prove that the full sequence (p_n) converges to p .⁷¹
- 3.21.** Prove the following analogue of Theorem 3.10(b): If (E_n) is a sequence of closed, nonempty, and bounded sets in a *complete* metric space X , if $E_n \supset E_{n+1}$, and if

$$\lim_{n \rightarrow \infty} \text{diam } E_n = 0,$$

then $\bigcap_{n=1}^{\infty} E_n$ consists of exactly one point.⁷²

72 Compactness weakened to closed-and-bounded, compensated by completeness: pick $x_n \in E_n$; shrinking diameters make (x_n) Cauchy, completeness gives the limit, closedness puts it in every E_n . Compare 2.36, where compactness did the work with no diameter hypothesis at all — and note that here, unlike there, “bounded” can even be dropped once diameters shrink.

73 The general Baire theorem, upgrading Exercise 2.30 from $\mathbb{R}^k$ to any complete metric space — with 3.21 replacing compactness as the engine. The construction: density of G_1 gives a ball inside it; density of G_2 gives a smaller ball inside both, radius halved; iterate, and 3.21 yields a point surviving every stage. This theorem underlies uniform boundedness and the open mapping theorem in functional analysis, and Exercise 2.30’s consequences (e.g. $\mathbb{R}$ uncountable) now hold in every complete metric space without isolated points.

74 Note where the limit lives: the real sequence $(d(p_n, q_n))$ is Cauchy, so it converges by 3.11(c) — completeness of $\mathbb{R}$ — even though (p_n) and (q_n) may converge to nothing in X . That asymmetry is exactly what the next exercise exploits.

- 3.22.** Suppose X is a nonempty complete metric space, and (G_n) is a sequence of dense open subsets of X . Prove Baire’s theorem, namely, that $\bigcap_{n=1}^{\infty} G_n$ is not empty. (In fact, it is dense in X .) *Hint:* Find a shrinking sequence of neighborhoods E_n such that $\overline{E_n} \subset G_n$, and apply Exercise 3.21.⁷³
- 3.23.** Suppose (p_n) and (q_n) are Cauchy sequences in a metric space X . Show that the sequence $(d(p_n, q_n))$ converges. *Hint:* For any m, n ,

$$d(p_n, q_n) \leq d(p_n, p_m) + d(p_m, q_m) + d(q_m, q_n);$$

it follows that

$$|d(p_n, q_n) - d(p_m, q_m)|$$

is small if m and n are large.⁷⁴

- 3.24.** Let X be a metric space.

(a) Call two Cauchy sequences $(p_n), (q_n)$ in X *equivalent* if

$$\lim_{n \rightarrow \infty} d(p_n, q_n) = 0.$$

Prove that this is an equivalence relation.

(b) Let X^* be the set of all equivalence classes so obtained. If $P \in X^*, Q \in X^*, (p_n) \in P, (q_n) \in Q$, define

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(p_n, q_n);$$

by Exercise 3.23, this limit exists. Show that the number $\Delta(P, Q)$ is unchanged if (p_n) and (q_n) are replaced by equivalent sequences, and hence that Δ is a distance function in X^* .

(c) Prove that the resulting metric space X^* is complete.

- (d) For each $p \in X$, there is a Cauchy sequence all of whose terms are p ; let P_p be the element of X^* which contains this sequence. Prove that

$$\Delta(P_p, P_q) = d(p, q)$$

for all $p, q \in X$. In other words, the mapping φ defined by $\varphi(p) = P_p$ is an *isometry* (i.e., a distance-preserving mapping) of X into X^* .

- (e) Prove that $\varphi(X)$ is dense in X^* , and that $\varphi(X) = X^*$ if X is complete. By (d), we may identify X and $\varphi(X)$ and thus regard X as embedded in the complete metric space X^* . We call X^* the *completion* of X .

75

- 3.25.** Let X be the metric space whose points are the rational numbers, with the metric $d(x, y) = |x - y|$. What is the completion of this space? (Compare Exercise 3.24.)⁷⁶

75 *The construction the appendix to Chapter 1 declined: a missing limit is represented by the sequences that reach for it, with two sequences identified when they reach for the same thing. This is Cantor's route to $\mathbb{R}$ and the one Tao's Analysis I takes as its foundation (his Chapter 5: reals are equivalence classes of rational Cauchy sequences). Note what makes it outrun Dedekind cuts, as the appendix's note anticipated: nothing here mentions order — only d — so every metric space acquires a completion, function spaces included. Part (c) has the construction's one subtlety: a Cauchy sequence of equivalence classes is unwound by picking representatives and a diagonal argument, the 3.7-style chase again.*

76 $\mathbb{R}$, of course — and the point of asking is that the completion was built without $\mathbb{R}$: 3.24 used only $\mathbb{Q}$'s own metric. Chapter 1 characterised $\mathbb{R}$ by order (least upper bounds); this exercise constructs it from distance. Same object, two foundations, and the uniqueness-up-to-isomorphism noted in Step 9 of the appendix is what guarantees they agree.

Appendix to Chapter 3

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Rudin's Chapter 3 is the most complete chapter in the book, and most of what the other texts add is not missing theory but missing *leverage*. He defines $\limsup$ as the supremum of the set of subsequential limits and never gives the formula one actually computes with. He proves the root and ratio tests and then remarks that both fail when the limit is 1 — without saying what to do next. He proves summation by parts but extracts no reusable bound from it. Each of those is one theorem away from being a tool.

What is here.

A. Upper and lower limits, finished. The inf-of-sup formula, the ε -characterisation, and the algebra — subadditivity, products, reciprocals — that Rudin's 3.19 only begins.

B. Past the root and ratio tests. Raabe's test, which compares against the p -series instead of the geometric series; the classical series that forces it; and the reason the hierarchy of tests can never terminate.

C. Summation by parts, quantitatively. Abel's lemma with an explicit constant, Abel's test with bounded rather than null weights, and the Dirichlet-kernel estimate that carries Chapter 3 into Fourier series.

D. Grouping and multiplying. What inserting parentheses is allowed to do, and the form of the product theorem that holds for *every* arrangement of the double array rather than the diagonal one.

E. Summation without an order. Rudin's 3.55 says an absolutely convergent series may be reordered freely. Tao takes that as licence to drop the order entirely and sum over an index set — after which double series and iterated sums are theorems rather than notation.

How to read it. Section A pays for itself immediately — read it with 3.16–3.19. Sections B–D are best read after 3.44 and 3.50 respectively, and E after 3.55. The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — UPPER AND LOWER LIMITS, FINISHED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Definition 3.16 puts E for the set of subsequential limits (in the extended reals) and defines $\limsup x_n = \sup E$. Theorem 3.17 then proves that $\limsup$ belongs to E and that no larger number does, and 3.19 gives the one comparison $a_n \leq b_n \Rightarrow \limsup a_n \leq \limsup b_n$.

The gap. That definition is superb for proving things and useless for computing. To find $\limsup$ from it you would first have to find every subsequential limit — which is harder than the original problem. The formula everyone actually uses, $\limsup x_n = \lim_n \sup_{k \geq n} x_k$, is never stated. Nor is the algebra: no subadditivity, no product rule, no reciprocal rule.

Where the gap bites. At 3.33 and 3.34, the root and ratio tests, which are stated in terms of $\limsup$ and therefore cannot be applied until one can evaluate a $\limsup$; at 3.39, the radius of convergence, where $1/\limsup \sqrt[n]{|c_n|}$ has to be computed for actual power series; and at 3.37, whose proof compares two upper limits and needs exactly the algebra that 3.19 does not supply.

What this section supplies. The monotone formula, with the proof that it agrees with Rudin's definition; the two-clause ε -characterisation that every later $\limsup$ argument runs on; and the algebra, including the counterexample showing why the inequalities are one-directional.

S3.1 Theorem. (*The formula.*) For any real sequence (x_n) ,

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k, \quad \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k,$$

the two limits existing in the extended reals because the sequences $M_n = \sup_{k \geq n} x_k$ and $m_n = \inf_{k \geq n} x_k$ are monotone.

SOURCE: ECNU 定理 7.9 (bk. p.158), which notes that many texts take this as the definition; Rudin never states it.

Proof. Since $\{x_k : k \geq n+1\} \subset \{x_k : k \geq n\}$, the suprema satisfy $M_{n+1} \leq M_n$: the sequence (M_n) decreases, so it has a limit M in the extended reals. Write E for Rudin's set of subsequential limits; we show $M = \sup E$.

If $L \in E$, say $x_{n_j} \rightarrow L$, then $x_{n_j} \leq M_n$ for every j with $n_j \geq n$, so $L \leq M_n$ for every n and hence $L \leq M$. Thus $\sup E \leq M$.

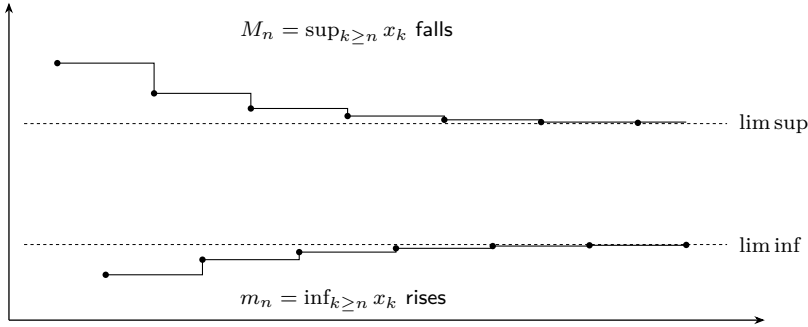
Conversely $M \in E$. Choose indices $n_1 < n_2 < \dots$ so that $x_{n_k} > M_{n_k} - 1/k$, which is possible because M_{n_k} is the supremum of the tail beginning at n_k . Since also $x_{n_k} \leq M_{n_k}$ and $M_{n_k} \rightarrow M$, we get $x_{n_k} \rightarrow M$. Hence $M \leq \sup E$. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** Two descriptions of one number are given, so the proof is two inequalities. Neither side can be computed, so each must be pinned by an approximation argument.
2. **Get monotonicity free.** Discarding terms can only lower a supremum. That single observation is what makes $\lim M_n$ exist at all — by monotone convergence (3.14), with no boundedness needed once $\pm\infty$ are allowed.
3. **One inequality by domination.** Every subsequential limit is trapped below every tail supremum, so $\sup E \leq M$. Nothing is constructed here.
4. **The other by construction.** To show $M \leq \sup E$ it is not enough that M bounds things — M must *be* a subsequential limit, so a subsequence converging to it has to be built. The recipe is the defining property of a supremum: at stage k pick a term within $1/k$ of M_{n_k} .
5. **Audit.** The least-upper-bound property is spent once, in step 4, to produce each x_{n_k} ; monotone convergence once, in step 2. Note which clause of Definition 1.8 does which job: “upper bound” gives step 3, “least” gives step 4 — the same division of labour as in Rudin’s 1.21.

WHY THIS IS THE VERSION YOU COMPUTE WITH

M_n is a decreasing sequence — taking the supremum over a smaller tail cannot increase it — and it is bounded below whenever (x_n) is. So the limit exists by monotone convergence, with no reference to subsequences at all. To evaluate $\limsup$ you now do two things you know how to do: take a supremum, then take a monotone limit. Rudin’s E -based definition and this one agree, and the agreement is worth seeing as two inequalities. Every subsequential limit is $\leq M_n$ for all n , hence $\leq \lim M_n$; and $\lim M_n$ is itself a subsequential limit, because for each n one can pick an index $k_n \geq n$ with $x_{k_n} > M_n - 1/n$. The second half is where the work is, and it is the same construction as ECNU’s proof that the largest cluster point exists.



The two staircases. Discarding a term can only lower a supremum and raise an infimum, so M_n decreases and m_n increases; their limits are the upper and lower limits. The picture also shows why the sequence converges exactly when the staircases meet.

S3.2 Theorem. (ε -characterisation.) $\bar{A} = \limsup x_n$ if and only if for every $\varepsilon > 0$: (i) $x_n < \bar{A} + \varepsilon$ for all sufficiently large n , and (ii) $x_n > \bar{A} - \varepsilon$ for infinitely many n . Equivalently: every $\alpha > \bar{A}$ is exceeded only finitely often, and every $\beta < \bar{A}$ is exceeded infinitely often.

SOURCE: ECNU 定理 7.7 and 7.7' (bk. p. 157). Rudin's 3.17(b) is condition (i) alone.

Proof. Let $\bar{A} = \limsup x_n$ and use S3.1. Given $\varepsilon > 0$, the numbers M_n decrease to $\bar{A}$, so $M_N < \bar{A} + \varepsilon$ for some N , and then $x_n \leq M_N < \bar{A} + \varepsilon$ for all $n \geq N$: that is (i). And $M_n \geq \bar{A}$ for every n , so $\sup_{k \geq n} x_k > \bar{A} - \varepsilon$, which produces some $k \geq n$ with $x_k > \bar{A} - \varepsilon$; as n is arbitrary there are infinitely many such k , which is (ii).

Conversely, suppose A satisfies (i) and (ii). By (i), $M_N \leq A + \varepsilon$ for some N , so $\bar{A} \leq A + \varepsilon$; by (ii), $M_n > A - \varepsilon$ for every n , so $\bar{A} \geq A - \varepsilon$. As ε was arbitrary, $\bar{A} = A$ by the ε -principle (S1.9). $\square$

Rudin's 3.17 says $\limsup$ belongs to E and that nothing above it does. The ε -form is the same content converted into the two statements one actually verifies.⁷⁷

S3.3 Theorem. (Algebra of upper and lower limits.) For bounded real sequences,

- (a) $\liminf x_n = -\limsup(-x_n)$;
- (b) $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$, with equality whenever $\lim x_n$ exists;
- (c) $\liminf x_n + \liminf y_n \leq \liminf(x_n + y_n)$;
- (d) if $x_n, y_n > 0$, then $(\liminf x_n)(\liminf y_n) \leq \liminf(x_n y_n)$;
- (e) if $\liminf x_n > 0$, then $\limsup(1/x_n) = 1/\liminf x_n$.

SOURCE: ECNU 定理 7.8, 例 2, and 习题 7.2 #2 (bk. pp. 158–159).

Proof. All five follow from S3.1 together with elementary facts about suprema over a common tail.

- (a) $\inf_{k \geq n}(-x_k) = -\sup_{k \geq n} x_k$; let $n \rightarrow \infty$.

⁷⁷ Read the pair as “eventually below, infinitely often nearly above.” Almost every $\limsup$ argument in later chapters — the root test at 3.33, the radius of convergence at 3.39 — uses exactly these two clauses, one to get an upper bound that eventually holds and one to find infinitely many terms witnessing sharpness.

(b) $\sup_{k \geq n}(x_k + y_k) \leq \sup_{k \geq n} x_k + \sup_{k \geq n} y_k$, since each term on the left is dominated by the right side; let $n \rightarrow \infty$. If $x_n \rightarrow x$ then applying the inequality to $y_n = (x_n + y_n) + (-x_n)$ gives $\limsup y_n \leq \limsup(x_n + y_n) + \limsup(-x_n) = \limsup(x_n + y_n) - x$, which is the reverse inequality.

(c) Apply (b) to $-x_n$ and $-y_n$ and use (a).

(d) For positive terms, $\inf_{k \geq n}(x_k y_k) \geq (\inf_{k \geq n} x_k)(\inf_{k \geq n} y_k)$; let $n \rightarrow \infty$.

(e) If $\liminf x_n > 0$ then $x_k > 0$ for large k and $\sup_{k \geq n}(1/x_k) = 1/\inf_{k \geq n} x_k$; let $n \rightarrow \infty$. $\square$

THE INEQUALITIES ARE ONE-DIRECTIONAL, AND THAT IS THE POINT

It is tempting to expect $\limsup(x_n + y_n) = \limsup x_n + \limsup y_n$. Take

$$x_n = (-1)^n, \quad y_n = (-1)^{n+1}.$$

Then $x_n + y_n = 0$ for every n , so the left side is 0, while the right side is $1 + 1 = 2$. The two sequences achieve their upper limits on *disjoint* sets of indices, and $\limsup$ cannot see that they never peak together. Equality is restored the moment one of the sequences converges, because a convergent sequence peaks everywhere at once. That is why Rudin can get away with 3.19 alone: he never adds two genuinely oscillating sequences. When you do — in the root test applied to a product, say — the inequality is all you have.

NOT IN RUDIN

SUPPLEMENT B — PAST THE ROOT AND RATIO TESTS

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorems 3.33 and 3.34 give the root and ratio tests; the Remark after 3.35 observes that both are silent when the relevant limit equals 1, illustrates with $\sum 1/n$ and $\sum 1/n^2$, and moves on.

The gap. The remark diagnoses a failure without saying what to do about it, and without explaining *why* the failure is unavoidable. A reader is left with the impression that the borderline is a small blind spot rather than the entrance to an infinite hierarchy.

Where the gap bites. On every series that matters at the boundary. Rudin's own 3.28 — $\sum 1/n^p$ — sits exactly there, and he must reach it by a separate device, the condensation test 3.27. The hypergeometric series, the central example of nineteenth-century analysis, is invisible to both of his tests. And

at 3.39 the radius of convergence of a power series is often decided precisely on the circle, where root and ratio give nothing.

What this section supplies. Raabe's test, which replaces the geometric yardstick by the p -series one and so resolves the whole borderline band; the classical series that forces it; and the reason no test can be last — there is no slowest convergent series, so Rudin's remark is a structural fact rather than an omission.

S3.4 Theorem. (*Raabe's test.*) Let $u_n > 0$. If there is $r > 1$ with

$$n\left(1 - \frac{u_{n+1}}{u_n}\right) \geq r \quad (n > N_0),$$

then $\sum u_n$ converges. If $n(1 - u_{n+1}/u_n) \leq 1$ for all $n > N_0$, then $\sum u_n$ diverges. In particular, if the limit $r = \lim n(1 - u_{n+1}/u_n)$ exists, then $r > 1$ gives convergence and $r < 1$ divergence.

SOURCE: ECNU 拉贝判别法 定理 12.10 (下 bk. pp. 14–15).

Proof. *Convergence.* Suppose $n(1 - u_{n+1}/u_n) \geq r > 1$ for $n > N_0$, so that $u_{n+1}/u_n \leq 1 - r/n$. Fix p with $1 < p < r$. Bernoulli's inequality $(1+t)^p \geq 1+pt$, applied with $t = -1/n$, gives

$$\left(\frac{n-1}{n}\right)^p \geq 1 - \frac{p}{n} \geq 1 - \frac{r}{n} \geq \frac{u_{n+1}}{u_n},$$

that is $n^p u_{n+1} \leq (n-1)^p u_n$. So the numbers $(n-1)^p u_n$ decrease for $n > N_0$ and are therefore bounded by some C , giving $u_n \leq C/(n-1)^p$. Since $p > 1$ the series $\sum (n-1)^{-p}$ converges (3.28), and the comparison test 3.25 finishes.

Divergence. If $n(1 - u_{n+1}/u_n) \leq 1$ then $u_{n+1}/u_n \geq 1 - 1/n = (n-1)/n$, so $nu_{n+1} \geq (n-1)u_n$ and the numbers $(n-1)u_n$ increase; hence $(n-1)u_n \geq c > 0$ and $u_n \geq c/(n-1)$. Since $\sum 1/(n-1)$ diverges, so does $\sum u_n$. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** The ratio test failed because it compares with a geometric series. So compare with a p -series instead — but the hypothesis is about *ratios*, so the comparison must be made between ratios rather than between terms.
2. **Compute the target ratio.** For $v_n = n^{-p}$ the ratio is $v_{n+1}/v_n = ((n)/(n+1))^p$, which Bernoulli bounds below by $1 - p/n$. So “ $u_{n+1}/u_n \leq 1 - p/n$ ” is exactly “the terms decay at least as fast as n^{-p} .”
3. **Buy room with $p < r$.** The hypothesis gives r ; the comparison needs a strict p above 1. Any p between them works, and this is the only place the strictness $r > 1$ is used.

- 4. Convert to a monotone quantity.** The ratio inequality says precisely that $(n-1)^p u_n$ decreases. A decreasing positive sequence is bounded, and boundedness is what the comparison test wants.
- 5. Audit.** Bernoulli is spent once, in step 2; the strict inequality $r > 1$ once, in step 3; the convergence of the p -series once, at the end. Run the same four steps with the inequalities reversed and $p = 1$ and the divergence half falls out — which is why the borderline case $r = 1$ is settled by neither.

WHAT CHANGED: THE YARDSTICK

The root and ratio tests compare $\sum u_n$ against a *geometric* series. When $u_{n+1}/u_n \rightarrow 1$ the comparison is vacuous, because the geometric series with ratio 1 is exactly the borderline.

Raabe compares against the p -series instead. Writing $u_{n+1}/u_n \approx 1 - r/n$ and comparing with $(1 - 1/n)^r$ shows that the quantity $n(1 - u_{n+1}/u_n)$ measures which power of n the terms decay like: $r > 1$ means faster than $1/n$, and $r < 1$ slower. So the test is “compare with $\sum n^{-r}$ ” with the comparison done through ratios rather than the terms themselves.

That reframing tells you immediately what the next rung must be. Since the p -series is itself borderline at $p = 1$, Raabe fails when $r = 1$ — and the next test compares against $\sum 1/(n(\log n)^p)$, which Rudin’s own 3.29 has already introduced.

The series that forces the issue, and why the ladder never ends

The standard example is

$$\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \right]^s,$$

whose ratio tends to 1 for *every* s , so ratio and root say nothing whatever. Raabe gives $r = s/2$, hence divergence for $s = 1$ and $s = 2$ and convergence for $s = 3$. The same computation settles the hypergeometric series $F(\alpha, \beta, \gamma, 1)$, which converges precisely when $\gamma > \alpha + \beta$ — historically the problem the whole hierarchy of tests was invented for.

ECNU then states the moral, and it is worth taking seriously: one can always build a finer test, and the process never terminates, because *there is no slowest convergent series*. Given any convergent $\sum a_n$ one can manufacture a convergent series whose terms dominate it eventually, so no single comparison series can serve as a universal boundary. Rudin’s 3.35 remark — that when the ratio is 1 “the series may converge or diverge” — is therefore not laziness but a structural fact.

The integral test, and why Rudin cannot state it here

Both Chinese texts prove: if f is nonnegative and decreasing on $[1, \infty)$, then $\sum f(n)$ and $\int_1^\infty f(x) dx$ converge together, via the sandwich $\sum_{k=2}^{n+1} f(k) \leq \int_1^{n+1} f \leq \sum_{k=1}^n f(k)$.

It is the fastest route to the p -series and to the iterated-logarithm ladder $\Sigma 1/(n \log n (\log \log n)^p)$. Rudin gets both by *Cauchy condensation* (3.27–3.29) instead, and he has no choice: the integral does not exist until Chapter 6, so a Chapter 3 integral test would be circular in his development. Come back to this box after 6.13, where the comparison becomes available — and notice that condensation and the integral test are the same idea, one summing over dyadic blocks and the other over unit intervals.

NOT IN RUDIN

SUPPLEMENT C — SUMMATION BY PARTS, QUANTITATIVELY

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorem 3.41 is the partial summation formula, and he uses it once, immediately, to prove 3.42 — if the partial sums of Σa_n are bounded and $b_n \downarrow 0$ then $\Sigma a_n b_n$ converges — from which 3.43 (alternating series) and 3.44 follow.

The gap. The formula is consumed on the spot. No *bound* is extracted from it, and no second theorem: Rudin has the Dirichlet test but not the Abel test, which trades the hypotheses the other way round.

Where the gap bites. At 3.44, where $\Sigma z^n/n^\alpha$ on $|z| = 1$ is handled by the Dirichlet route because the Abel route is unavailable; and, more seriously, in Chapter 7, where the uniform versions of these tests are wanted and a uniform conclusion needs an explicit constant, not merely a convergence statement.

What this section supplies. The bound — Abel's lemma, with the constant 3 — from which Rudin's 3.42 and its missing twin both follow in a line; and the Dirichlet kernel estimate, which is the standard bridge from this chapter to Fourier series and which 3.44 stops just short of.

S3.5 Theorem. (*Abel's lemma.*) Let (ε_k) be monotone, let $\sigma_k = v_1 + \cdots + v_k$, and suppose $|\sigma_k| \leq A$ for $k \leq n$. Put $\varepsilon = \max_k |\varepsilon_k|$. Then

$$\left| \sum_{k=1}^n \varepsilon_k v_k \right| \leq 3\varepsilon A.$$

Proof. Partial summation (3.41) gives

$$\sum_{k=1}^n \varepsilon_k v_k = \sum_{k=1}^{n-1} (\varepsilon_k - \varepsilon_{k+1}) \sigma_k + \varepsilon_n \sigma_n.$$

Because (ε_k) is monotone, the differences $\varepsilon_k - \varepsilon_{k+1}$ all have one sign, so

$$\sum_{k=1}^{n-1} |\varepsilon_k - \varepsilon_{k+1}| = |\varepsilon_1 - \varepsilon_n| \leq 2\varepsilon.$$

Bounding every $|\sigma_k|$ by A therefore gives $|\sum \varepsilon_k v_k| \leq 2\varepsilon A + \varepsilon A = 3\varepsilon A$.
□

The proof is Rudin's 3.41 followed by one observation: because (ε_k) is monotone, the differences $\varepsilon_k - \varepsilon_{k+1}$ all have the same sign, so $\sum_k |\varepsilon_k - \varepsilon_{k+1}| = |\varepsilon_1 - \varepsilon_n| \leq 2\varepsilon$, and the boundary terms contribute εA more.⁷⁸

S3.6 Theorem. (*Abel's test.*) If (a_n) is monotone and **bounded** — not necessarily null — and $\sum b_n$ converges, then $\sum a_n b_n$ converges.

SOURCE: ECNU 定理 12.15 (下 bk. p. 22).

Proof. Being monotone and bounded, (a_n) converges, say to a ; put $b_n = a_n - a$, so (b_n) is monotone with $b_n \rightarrow 0$. Then

$$\sum a_n c_n = a \sum c_n + \sum b_n c_n.$$

The first series converges by hypothesis. For the second, the partial sums of $\sum c_n$ are bounded — a convergent sequence is bounded — and $b_n \downarrow 0$, so Rudin's 3.42 applies. Hence $\sum a_n c_n$ converges. □

Two tests, and which hypothesis each spends

Rudin's 3.42 is the *Dirichlet* test: partial sums of $\sum b_n$ merely bounded, but $a_n \downarrow 0$. Abel's test trades those hypotheses: $\sum b_n$ must actually converge, but (a_n) need only be monotone and bounded. Neither implies the other, and both fall out of Abel's lemma.

	Dirichlet (Rudin 3.42)	Abel (S3.6)
on $\sum b_n$	partial sums bounded	converges
on (a_n)	monotone, $\rightarrow 0$	monotone, bounded
typical use	$\sum \frac{\sin nx}{n}$	multiplying a convergent series by a damping factor

Rudin never states Abel's form, and its absence is felt at 3.44, where he handles $\sum z^n/n^\alpha$ on $|z| = 1$ by the Dirichlet route.

S3.7 Theorem. (*The Dirichlet kernel bound.*) For $x \in (0, 2\pi)$,

$$\frac{1}{2} + \sum_{k=1}^n \cos kx = \frac{\sin((n + \frac{1}{2})x)}{2 \sin(x/2)},$$

⁷⁸ This is exactly the telescope Rudin performs inside the proof of 3.42, extracted and given a name. Rudin's version is consumed on the spot to prove one convergence theorem; as a standalone inequality it is reusable, and it is what the uniform-convergence versions in Chapter 7 run on.

so the partial sums of $\sum \cos kx$ are bounded on any interval $[\delta, 2\pi - \delta]$. Consequently, if $a_1 \geq a_2 \geq \cdots$ with $a_n \rightarrow 0$, then $\sum a_n \cos nx$ and $\sum a_n \sin nx$ converge for every $x \in (0, 2\pi)$.

SOURCE: ECNU 例 3 (下 bk. pp. 22–23).

Proof. For $x \in (0, 2\pi)$ we have $\sin(x/2) \neq 0$. The product formula $2 \cos kx \sin(x/2) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x)$ makes the sum telescope:

$$2 \sin \frac{x}{2} \left[\frac{1}{2} + \sum_{k=1}^n \cos kx \right] = \sin \frac{x}{2} + \sin((n + \frac{1}{2})x) - \sin \frac{x}{2} = \sin((n + \frac{1}{2})x),$$

which is the identity. Hence $|\sum_{k \leq n} \cos kx| \leq 1/(2|\sin(x/2)|) + \frac{1}{2}$, a bound independent of n and finite for each fixed $x \in (0, 2\pi)$. With $a_n \downarrow 0$, Rudin's 3.42 gives convergence of $\sum a_n \cos nx$; the same telescoping with $\sin$ in place of $\cos$ bounds the partial sums of $\sum \sin kx$ and settles that series too. $\square$

This is the standard bridge from Chapter 3 to Fourier series: it makes $\sum(\sin nx)/n$ and $\sum(\cos nx)/n$ convergent everywhere on $(0, 2\pi)$, none of which follows from absolute convergence, since $\sum 1/n$ diverges.⁷⁹

NOT IN RUDIN

SUPPLEMENT D — GROUPING AND MULTIPLYING

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Definition 3.48 defines the product of two series by the anti-diagonals $c_n = \sum_k a_k b_{n-k}$, motivated by power series; 3.49 shows the product of two conditionally convergent series can diverge; 3.50 (Mertens) proves convergence to AB when one factor is absolutely convergent.

The gap. Two operations on series are performed silently throughout the chapter and never licensed. *Grouping* — inserting parentheses — is used inside the condensation proof at 3.27 and in the Remark after 3.44, but never stated as a theorem, so a reader has no way to know it is legal while rearrangement is not. And the choice of the anti-diagonal ordering at 3.48 is presented as forced by power series, leaving open what happens under any other arrangement of the same products.

Where the gap bites. At 3.49 especially, where the divergence of a product looks like a pathology of multiplication rather than the price of insisting on one particular traversal of a doubly-indexed array.

What this section supplies. The grouping theorem with its one-directional caveat, so that the legal manipulation is cleanly

⁷⁹ The identity is proved by multiplying the left side by $2 \sin(x/2)$ and telescoping the products $2 \cos kx \sin(x/2) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x)$. Note where the bound degrades: the denominator $2 \sin(x/2)$ tends to 0 as $x \rightarrow 0$, so the partial sums are bounded on each $[\delta, 2\pi - \delta]$ but not uniformly on $(0, 2\pi)$ — which is exactly the phenomenon Chapter 7 will call non-uniform convergence.

separated from the illegal one; and Cauchy's theorem, that under absolute convergence *every* arrangement of the array converges to AB — which puts 3.50 and 3.49 in their proper places.

S3.8 Theorem. (*Grouping.*) *Inserting parentheses in a convergent series — that is, passing to any subsequence of the partial sums by grouping consecutive terms — changes neither convergence nor the sum.*

SOURCE: ECNU 定理 12.4 (下 bk. pp. 4–5).

Proof. Let s_n be the partial sums of the original series and let the parentheses end at indices $n_1 < n_2 < \dots$. The partial sums of the grouped series are exactly $s_{n_1}, s_{n_2}, \dots$ — a subsequence of (s_n) . A subsequence of a convergent sequence converges to the same limit (3.6(a)). $\square$

GROUPING IS ONE-DIRECTIONAL, UNLIKE REARRANGEMENT

The grouped partial sums form a *subsequence* of (s_n) , and a subsequence of a convergent sequence converges to the same limit — that is the whole proof. The converse fails:

$$(1 - 1) + (1 - 1) + (1 - 1) + \dots = 0$$

converges while $1 - 1 + 1 - 1 + \dots$ diverges. So parentheses may be *inserted* into a convergent series but not *removed* from one.

Keep this distinct from the rearrangement theorem 3.54. Rearranging changes the order of the terms and can change the sum of a conditionally convergent series to anything at all; grouping preserves the order and is always safe in the direction stated. Rudin never states the grouping theorem, although he uses it — in the condensation proof at 3.27 and in the Remark after 3.44.

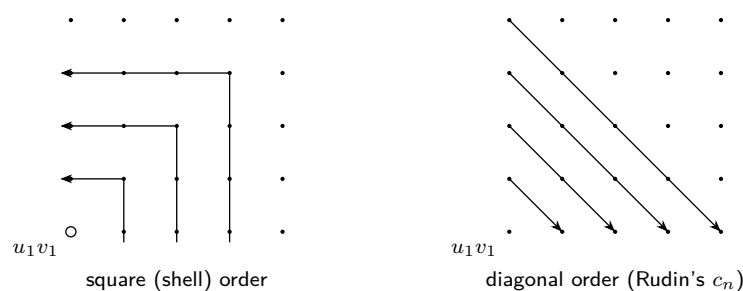
S3.9 Theorem. (*Cauchy's product theorem, any arrangement.*) *If $\Sigma u_n = A$ and $\Sigma v_n = B$ both converge absolutely, then the series formed from all products $u_i v_j$, each taken once, **in any order whatever**, converges absolutely to AB .*

SOURCE: ECNU 柯西定理 12.14 (下 bk. pp. 19–21).

Proof. Let (w_k) enumerate the products $u_i v_j$ in any order. Any finite collection of them involves indices $i, j \leq N$ for some N , so

$$\sum_{k \leq K} |w_k| \leq \left(\sum_{i \leq N} |u_i| \right) \left(\sum_{j \leq N} |v_j| \right) \leq \left(\sum_i |u_i| \right) \left(\sum_j |v_j| \right).$$

The partial sums of $\Sigma |w_k|$ are bounded, so Σw_k converges absolutely; by 3.55 its sum is the same for every enumeration. Evaluate it along the square ordering, in which the partial sum after the N th shell is exactly $A_N B_N$, and $A_N B_N \rightarrow AB$ by 3.3(c). $\square$



Two ways to enumerate the same array of products $u_i v_j$. Along shells the partial sums are exactly $A_n B_n$, which is why the square order proves the theorem; along anti-diagonals they are Rudin's C_n , which is the order that makes the product a series in the sense of 3.48. Absolute convergence is what makes the two agree.

How this relates to Mertens (3.50)

The two theorems point in opposite directions. Rudin's 3.50 fixes the diagonal ordering — the one that matters for power series — and weakens the hypothesis: only *one* factor need converge absolutely. S3.9 keeps the stronger hypothesis, both absolutely convergent, and buys freedom of arrangement.

The proof of S3.9 is the cleaner of the two, because the square ordering makes the partial sums *exactly* $A_n B_n$ rather than approximately so; all the difficulty in Mertens' theorem comes from the diagonal ordering cutting the array along a line the factors do not respect. Seen this way, Example 3.49 — where the Cauchy product of two conditionally convergent series diverges — is not a pathology but the expected price of insisting on the diagonal.

NOT IN RUDIN

SUPPLEMENT E — SUMMATION WITHOUT AN ORDER

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 3

What Rudin does. Theorem 3.54 shows a conditionally convergent series can be rearranged to any prescribed value, and 3.55 shows an absolutely convergent one cannot: every rearrangement has the same sum.

The gap. 3.55 is stated as a stability property — the sum survives reordering — rather than as what it actually is: a licence to forget the ordering altogether. Rudin never draws that conclusion, so a series remains, for him, a function on $\mathbb{N}$, and anything indexed differently must be converted to one by hand.

Where the gap bites. At 3.48, where the product of two series has to be *defined* by choosing a traversal of a square array, and at 3.49, where the choice turns out to matter. Later, in Chapter 8, power series are multiplied and rearranged repeatedly, and

each manipulation is justified afresh.

What this section supplies. Tao's reading of 3.55: for absolutely summable families the sum is a function of the *index set*, so it may be defined with no order at all. Double series and iterated sums then become theorems — Fubini for series — rather than notational accidents.

S3.10 Definition. (*Sums over arbitrary index sets.*) Let X be a set and $f : X \rightarrow \mathbb{R}$. If X is countable, choose a bijection $g : \mathbb{N} \rightarrow X$ and put $\sum_{x \in X} f(x) = \sum_{n=0}^{\infty} f(g(n))$, provided the latter converges absolutely; the value does not depend on g . If X is uncountable, the same definition applies once one knows that $\{x : f(x) \neq 0\}$ is at most countable.

SOURCE: Tao I, §8.2, Def. 8.2.1–8.2.4 and Lemma 8.2.5 (bk. pp. 187–189).

WHAT RUDIN'S 3.55 BECOMES

Rudin proves that an absolutely convergent series may be rearranged freely (3.55) and that a conditionally convergent one may be rearranged to any sum whatever (3.54). Tao reads the first as a licence: since the value is independent of the enumeration, *the ordering was never part of the data*. A series is then a function on an index set, and $\sum_{x \in X} f(x)$ is well defined with no order in sight. The dividend is that families indexed by things other than $\mathbb{N}$ — pairs, grids, arbitrary countable sets — need no separate theory. Rudin has to manufacture an ordering to define the Cauchy product at 3.48 (his c_n are the anti-diagonals), and Example 3.49's failure is precisely the price of that manufactured ordering.

S3.11 Lemma. If $\sum_{x \in X} f(x)$ is absolutely convergent, then $\{x \in X : f(x) \neq 0\}$ is at most countable — however large X is.

SOURCE: Tao I, Lemma 8.2.5 (bk. p. 188).

So an absolutely summable family over an uncountable index set is really a series in disguise.⁸⁰

S3.12 Theorem. (*Fubini for series.*) If $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ has $\sum_{(n,m)} f(n,m)$ absolutely convergent, then

$$\sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} f(n,m) \right) = \sum_{(n,m) \in \mathbb{N} \times \mathbb{N}} f(n,m) = \sum_{m=0}^{\infty} \left(\sum_{n=0}^{\infty} f(n,m) \right).$$

SOURCE: Tao I, Thm 8.2.2 (bk. p. 188). Rudin states nothing about iterated sums in Chapter 3.

Proof. Write S for the unordered sum, which exists by S3.10. Given $\varepsilon > 0$, absolute convergence supplies a finite set $F \subset \mathbb{N} \times \mathbb{N}$ with $\sum_{(n,m) \notin F} |f(n,m)| < \varepsilon$; choose N with $F \subset [0, N]^2$.

Each inner sum $g(n) = \sum_m f(n,m)$ converges absolutely, and $\sum_{n > N} |g(n)| \leq \sum_{n > N} \sum_m |f(n,m)| < \varepsilon$. Comparing $\sum_{n \leq N} g(n)$ with

⁸⁰ The proof is the standard countability argument in Rudin's own idiom: for each n the set where $|f| > 1/n$ must be finite, or the sum would exceed any bound; a countable union of finite sets is countable. Compare Exercise S2.18 in the Chapter 2 appendix — disjoint intervals are countable for the same reason.

S , the terms they fail to share all lie outside F , so the two differ by at most ε . Hence $|\sum_{n \leq N} g(n) - S| \leq \varepsilon$ and, adding the tail estimate, $|\sum_n g(n) - S| \leq 2\varepsilon$. As ε was arbitrary, $\sum_n g(n) = S$; the argument with the roles of n and m exchanged gives the other iterated sum. $\square$

Why this is the theorem the Cauchy product wanted

S3.9 said that an absolutely convergent double array may be summed in any arrangement. S3.12 says the two most useful arrangements — by rows and by columns — also agree with each other and with the unordered sum. Together they are the complete story of the array $u_i v_j$, of which Rudin's 3.50 is the single diagonal slice.

Absolute convergence is not decoration here. The standard counterexample sets $f(n, n) = 1$, $f(n, n+1) = -1$, and $f = 0$ elsewhere: summing along rows gives 0 every time, hence 0; summing along columns gives 1 then 0 forever, hence 1. Nothing is infinite or badly behaved — every row and every column is a finite sum — and still the two iterated sums differ, because the family is not absolutely summable. Rudin's Exercise 3.25 asks for this phenomenon in a different guise.

Chapter 8 will need this theorem to multiply and rearrange power series, and Chapter 11 will need its integral analogue.

S3.13 Theorem. (*Means, and the ratio-to-root corollary.*) Let $a_n \rightarrow a$. Then

(a) $\frac{a_1 + \cdots + a_n}{n} \rightarrow a$ (Cesàro);

(b) if moreover $a_n > 0$, then $\sqrt[n]{a_1 a_2 \cdots a_n} \rightarrow a$;

(c) consequently, if $b_n > 0$ and $b_{n+1}/b_n \rightarrow a$, then $\sqrt[n]{b_n} \rightarrow a$.

SOURCE: ECNU 总练习题 2 #3–#4 (bk. p. 39); USTC Ch. 1 suppl. #6–#11 (bk. pp. 38–39).

Proof. (a) Given $\varepsilon > 0$ choose N with $|a_k - a| < \varepsilon/2$ for $k > N$. For $n > N$,

$$\left| \frac{1}{n} \sum_{k \leq n} a_k - a \right| \leq \frac{1}{n} \sum_{k \leq N} |a_k - a| + \frac{\varepsilon}{2},$$

and the first term tends to 0 as $n \rightarrow \infty$ with N fixed, so it is below $\varepsilon/2$ eventually.

(b) If $a > 0$, apply (a) to $\log a_n$ and exponentiate, using continuity of $\exp$. If $a = 0$, given $\varepsilon > 0$ we have $0 < a_k < \varepsilon$ for $k > N$, and the geometric mean is at most $(\prod_{k \leq N} a_k)^{1/n} \varepsilon^{(n-N)/n} \rightarrow \varepsilon$.

(c) Put $a_1 = b_1$ and $a_k = b_k/b_{k-1}$ for $k \geq 2$, so that $a_1 a_2 \cdots a_n = b_n$ and $a_n \rightarrow a$. Now (b) gives $\sqrt[n]{b_n} \rightarrow a$. $\square$

The corollary that explains Rudin's 3.37

Part (c) is the exact statement behind Rudin's Theorem 3.37,

$$\liminf \frac{c_{n+1}}{c_n} \leq \liminf \sqrt[n]{c_n} \leq \limsup \sqrt[n]{c_n} \leq \limsup \frac{c_{n+1}}{c_n},$$

whose corollary he states as “the root test is stronger than the ratio test.” Read through (c): the root test sees a *geometric mean* of the ratios, and geometric means are gentler than the sequence they average — they converge whenever it does, and sometimes when it does not. Rudin's Example 3.35, where the ratios oscillate between 2 and 1/8 while the roots converge, is exactly a case where the mean is tamer than its terms.

Applying (c) to $b_n = n^n/n!$ gives $\lim n/\sqrt[n]{n!} = e$ in one line, and (a) to $\sqrt[n]{n}$ gives $\lim(1 + \sqrt{2} + \sqrt[3]{3} + \cdots + \sqrt[n]{n})/n = 1$. Rudin has the Cesàro mean only as Exercise 3.14(a) and never states the geometric version.

ONE THEOREM YOU WILL LOOK FOR AND NOT FIND

The Stolz–Cesàro theorem — the discrete l'Hospital rule, that $\lim(a_n - a_{n-1})/(b_n - b_{n-1}) = L$ forces $\lim a_n/b_n = L$ for (b_n) strictly increasing and unbounded — appears **in none of the four comparison texts**, and not in Rudin. All four give only the Cesàro special cases of S3.13. The continuous analogue does appear, as Yu Pin's problem A9* (p. 127), where it is attributed to Cauchy rather than to Stolz; it is exercise S4.3 in the Chapter 4 appendix. If you want the discrete theorem you will have to prove it, and S3.13(a) is the case $b_n = n$.

NOT IN RUDIN**APPENDIX EXERCISES****HOW THESE DIFFER FROM RUDIN'S TWENTY-FIVE**

Rudin's own Exercises 3.1–3.25 end the chapter proper. These eighteen come from the four comparison texts.

Group I (1–5): sequences. Coupled recursions — Gauss's arithmetic–geometric mean among them — where Rudin's 3.16–3.17 use a single recursion; plus the means package and bounded variation.

Group II (6–12): series. The decay rate of the terms of a convergent series, Cauchy–Schwarz in ℓ^2 , infinite products, and the positive/negative split that Rudin buries inside the proof of 3.54.

Group III (13–16): upper and lower limits. The algebra of Supplement A, put to work.

Group IV (17–18): summation over sets. Iterated sums, and the counterexample that shows absolute convergence is not decoration.

Marked ★: harder, or requiring a result from a later chapter.

S3.1. (*Arithmetic–geometric mean.*) Let $a_1 > b_1 > 0$ and put

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \sqrt{a_n b_n}.$$

Prove that $\lim a_n$ and $\lim b_n$ both exist and are equal.⁸¹

SOURCE: ECNU 习题 2.3 #11 (bk. p. 38).

S3.2. (*Arithmetic–harmonic mean.*) With $a_1, b_1 > 0$ put $a_{n+1} = (a_n + b_n)/2$ and $b_{n+1} = 2a_n b_n / (a_n + b_n)$. Prove both limits exist and equal $\sqrt{a_1 b_1}$.⁸²

SOURCE: ECNU 总练习题 2 #8 (bk. p. 39).

S3.3. Prove that $a_n \rightarrow a$ implies $(a_1 + \cdots + a_n)/n \rightarrow a$, and that if also $a_n > 0$ then $\sqrt[n]{a_1 \cdots a_n} \rightarrow a$. Deduce that $b_{n+1}/b_n \rightarrow a$ implies $\sqrt[n]{b_n} \rightarrow a$ for $b_n > 0$, and use it to evaluate $\lim n/\sqrt[n]{n!}$.⁸³

SOURCE: ECNU 总练习题 2 #3–#4 (bk. p. 39); USTC Ch. 1 suppl. #6–#11.

S3.4. Suppose $\sum_{k=1}^n |a_{k+1} - a_k| \leq M$ for every n . Prove that (a_n) converges.⁸⁴

SOURCE: ECNU 总练习题 2 #6 (bk. p. 39); USTC Ch. 1 suppl. #5.

S3.5. Prove that if every convergent subsequence of a bounded sequence has the same limit L , then the sequence itself converges to L . Deduce that a bounded divergent sequence has two subsequences with different limits.⁸⁵

SOURCE: USTC §1.2 (bk. p. 15), left there as an exercise.

S3.6. Prove that if $\sum u_n$ is a convergent series of positive terms and (u_n) is monotone, then $\lim n u_n = 0$.⁸⁶

SOURCE: ECNU 总练习题 12 #1 (下 bk. p. 24).

S3.7. Prove that if $\sum a_n^2$ and $\sum b_n^2$ converge then $\sum a_n b_n$ and $\sum (a_n + b_n)^2$ converge, and

$$\left(\sum a_n b_n \right)^2 \leq \sum a_n^2 \sum b_n^2,$$

$$\left(\sum (a_n + b_n)^2 \right)^{1/2} \leq \left(\sum a_n^2 \right)^{1/2} + \left(\sum b_n^2 \right)^{1/2}.$$

81 Gauss's AGM. The engine is the inequality $\sqrt{xy} \leq (x+y)/2$, which gives $b_n < b_{n+1} < a_{n+1} < a_n$: the two sequences are monotone in opposite directions and interleaved, so 3.14 applies to each and the gap $a_{n+1} - b_{n+1} \leq (a_n - b_n)/2$ closes. Rudin's Exercises 3.16–3.17 are single recursions converging to a square root; a coupled recursion is materially different, and the common limit here has no elementary closed form — it is an elliptic integral.

82 The companion to exercise 1, and this time the answer is elementary — because $a_n b_n$ is invariant: $a_{n+1} b_{n+1} = a_n b_n$ for every n . Spot the invariant and the problem collapses. Compare the strategy index's second pattern in Chapter 1: hunt for the quantity the recursion does not move.

83 Rudin gives only the first part, as Exercise 3.14(a). The deduction is the content of 3.37 — the root test dominates the ratio test — and the last limit is e , obtained by applying (c) to $b_n = n^n/n!$. Show also that the converse fails: averaging destroys information, so $(-1)^n$ has Cesàro mean 0 without converging.

84 Bounded variation implies convergence — the sequence version of a notion Chapter 6 will develop for functions. The proof is one line once you see it: $a_n = a_1 + \sum_{k < n} (a_{k+1} - a_k)$ exhibits (a_n) as the partial sums of an absolutely convergent series. Note the converse fails: a convergent sequence can wander infinitely far, as $a_n = \sum_{k \leq n} (-1)^k/k$ shows.

85 The cleanest packaging of Rudin's 3.6–3.7. Argue by contradiction: if $x_n \not\rightarrow L$, infinitely many terms stay ε away, and that subsequence is bounded, so by 3.6(b) it has a convergent sub-subsequence — whose limit is not L . Note the hypothesis bounded is essential: $x_n = n$ has no convergent subsequence at all, so the condition holds vacuously.

86 Sharper than 3.23, which gives only $u_n \rightarrow 0$. The trick is to look at a block: monotonicity makes

$u_{n+1} + \cdots + u_{2n} \geq nu_{2n}$, and the left side tends to 0 by the Cauchy criterion. Note what fails without monotonicity — one can have a convergent series with $nu_n \not\rightarrow 0$ by putting spikes at sparse indices. This is the quantitative reason the harmonic series is the boundary.

87 Cauchy–Schwarz and Minkowski in ℓ^2 — Theorem 1.35 and 1.37(e) in infinite dimensions, which Rudin never revisits after Chapter 1. Prove the finite versions and pass to the limit; the only care needed is that the partial sums are increasing and bounded. This is the first appearance of the space ℓ^2 , which Chapter 11 treats seriously.

88 Infinite products reduced entirely to Chapter 3 by taking logarithms: convergence of $\sum \log(1 + a_k)$ is equivalent to convergence of $\sum a_k$, since $\log(1 + x) \sim x$. The evaluation telescopes — multiply by $(1 - q)$ and watch

$(1 - q)(1 + q)(1 + q^2)(1 + q^4) \cdots = 1 - q^{2^n}$ — giving $1/(1 - q)$. None of the four texts devotes a section to infinite products; this exercise is the whole theory in miniature.

89 Two decay-rate arguments. The first is comparison with $\sum 1/n^2$; the second is the arithmetic–geometric mean inequality, $\sqrt{u_n u_{n+1}} \leq (u_n + u_{n+1})/2$. Both illustrate the same habit: when a series resists direct attack, bound its terms by something already known to converge, and prefer the crudest bound that works.

90 The lemma inside Rudin’s proof of 3.54, isolated. If both converged, $\sum |u_n| = \sum(p_n + q_n)$ would converge, contradicting conditionality; if exactly one converged, $\sum u_n = \sum(p_n - q_n)$ would diverge. So both piles are infinite — which is precisely the fuel Riemann’s rearrangement burns. Worth extracting because the statement, not the rearrangement, is what makes 3.54 believable.

87

SOURCE: ECNU 总练习题 12 #8 (下 bk. p. 24).

S3.8. Let $a_n > 0$. Prove that $\prod_{k=1}^n (1 + a_k)$ converges (to a finite nonzero limit) if and only if $\sum a_n$ converges. Evaluate $\prod_{k=0}^{\infty} (1 + q^{2^k})$ for $|q| < 1$.⁸⁸

SOURCE: ECNU 习题 12.2 #15 (下 bk. p. 17); USTC 习题 1.1 #8(5).

S3.9. Let $a_n \geq 0$ with (na_n) bounded. Prove $\sum a_n^2$ converges. Prove also that if $\sum u_n$ is a convergent positive series then $\sum \sqrt{u_n u_{n+1}}$ converges.⁸⁹

SOURCE: ECNU 习题 12.2 #5, #7 (下 bk. p. 16).

S3.10. With $p_n = (|u_n| + u_n)/2$ and $q_n = (|u_n| - u_n)/2$, prove that if $\sum u_n$ converges conditionally then $\sum p_n$ and $\sum q_n$ both diverge.⁹⁰

SOURCE: ECNU 习题 12.3 #4 (下 bk. p. 23).

S3.11. ★ Prove that $\sum_{n=1}^{\infty} \frac{(-1)^{\lfloor \sqrt{n} \rfloor}}{n}$ converges.⁹¹

SOURCE: ECNU 习题 12.3 #8 (下 bk. p. 24).

S3.12. Given a convergent series of positive terms $\sum u_n$, must there exist $\varepsilon > 0$ with $u_n/n^{-(1+\varepsilon)} \rightarrow c$ for some $c > 0$?⁹²

SOURCE: ECNU 总练习题 12 #5(3) (下 bk. p. 24).

S3.13. Prove the parts of S3.3: $\liminf a_n = -\limsup(-a_n)$; $\liminf a_n + \liminf b_n \leq \liminf(a_n + b_n)$; and, for positive sequences with $\liminf a_n > 0$, $\limsup(1/a_n) = 1/\liminf a_n$.⁹³

SOURCE: ECNU 习题 7.2 #2 (bk. p. 159).

S3.14. Let $a_n > 0$ and suppose $\limsup a_n \cdot \limsup(1/a_n) = 1$. Prove that (a_n) converges.⁹⁴

SOURCE: ECNU 习题 7.2 #4 (bk. p. 159).

S3.15. ★ Suppose $\liminf x_n = A < B = \limsup x_n$ and $\lim(x_{n+1} - x_n) = 0$. Prove that the set of subsequential limits of (x_n) is exactly $[A, B]$.⁹⁵

SOURCE: ECNU 总练习题 7 #3 (bk. p. 160), starred there as well.

S3.16. Show by example that $\limsup(a_n b_n) = \limsup a_n \cdot \limsup b_n$ can fail for positive bounded sequences, and prove it holds when one of the sequences converges.⁹⁶

SOURCE: ECNU 习题 7.2 #2 (bk. p. 159), product part.

S3.17. Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ by $f(n, n) = 1$, $f(n, n+1) = -1$, and $f = 0$ otherwise. Compute both iterated sums $\sum_n (\sum_m f(n, m))$ and $\sum_m (\sum_n f(n, m))$, and explain why S3.12 does not apply.⁹⁷

SOURCE: Tao I, §8.2 discussion (bk. p. 188).

S3.18. ★ Let X be any set and $f : X \rightarrow \mathbb{R}$ with $\sum_{x \in X} |f(x)|$ finite. Prove that $\{x : f(x) \neq 0\}$ is at most countable, and that $\sum_{x \in X} f(x)$ is independent of the enumeration used to compute it.⁹⁸

SOURCE: Tao I, Lemma 8.2.5 and Ex. 8.2.2 (bk. p. 188).

91 The signs come in runs of lengths 3, 5, 7, ..., so the alternating series test (3.43) does not apply at all. Group the terms into blocks of constant sign, show the grouped series converges by comparison, and then use the fact that the terms tend to 0 to remove the parentheses — which is legitimate here precisely because the block lengths grow slowly. A genuine stress test of S3.5 and S3.8.

92 No — and the exercise is the concrete form of “there is no slowest convergent series,” the moral of Supplement B. Convergent series can decay arbitrarily slowly, so no single power of n serves as a universal gauge. This is why the Raabe/Gauss hierarchy has no last member, and why Rudin’s remark after 3.35 is a structural fact rather than an omission.

93 All three fall out of the inf-of-sup formula S3.1 plus the elementary facts $\sup(-A) = -\inf A$ and $\inf(A + B) \geq \inf A + \inf B$. Doing them once makes clear why the inequalities point the way they do: taking a supremum over a tail is a sublinear operation, and sublinearity is one-directional.

94 A one-line consequence of S3.3(e) once you see it: the hypothesis says $\limsup a_n = \liminf a_n$, and 3.17 finishes it. The pleasure is that a statement about products of upper limits secretly encodes “the two limits coincide.” Note the hypothesis forces $\liminf a_n > 0$, so the reciprocal rule applies.

95 A beautiful result with no analogue anywhere in Rudin. The hypothesis says the sequence moves in ever-smaller steps while still travelling between A and B infinitely often — so it cannot skip any value in between. Formally: given $c \in (A, B)$ and $\varepsilon > 0$, find a stretch of indices crossing from below c to above c ; because the steps are eventually smaller than ε , some term of that stretch lands within ε of c . Compare S2.10 in the Chapter 2 appendix, where consecutive-terms-close fails to give convergence: here the same hypothesis is turned into positive information.

96 *The multiplicative twin of the pitfall at S3.3. Take a_n alternating between 1 and 2 and b_n alternating the other way: each has $\limsup = 2$, but the product is constantly 2. The repair is the same — a convergent factor peaks everywhere at once, so it cannot dodge the other sequence's peaks.*

97 *The answers are 0 and 1. Every row and every column is a finite sum, so nothing diverges and nothing is subtle — and still the order of summation matters, because $\sum_{(n,m)} |f(n,m)|$ is infinite. The lesson is that absolute convergence is the entire hypothesis of Fubini, not a technical convenience. Rudin's Exercise 3.25 asks for a relative of this.*

98 *S3.11 and the well-definedness behind S3.10. For the first part, show $\{x : |f(x)| > 1/n\}$ is finite for each n and take the union. For the second, the two enumerations differ by a bijection of $\mathbb{N}$, so this is Rudin's 3.55 — which is exactly the point: absolute convergence is what makes “the sum over a set” meaningful, and 3.54 says the notion collapses without it.*